Status Paper on Sugarcane

Directorate of Sugarcane Development

सत्यमेव जयते

Govt. of India

Ministry of Agriculture

8th Floor, Kendriya Bhawan, Aliganj,

Lucknow-226024 (U.P.)

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FOREWORD
COLOUR PAGE

PREFACE
COLOUR PAGE

CONTENT

ANNEXURES

S.No.	Particular
I(a)	State wise area of sugarcane crop in India
I(b)	State wise production of sugarcane crop in India
I(c)	State wise productivity of sugarcane crop in India
II	State wise sugar production during last 10 years
III	Area, Production and Yield of Sugarcane in India
IV	Area, production and yield of sugarcane in major growing countries (2006-2010)
V	Gap of yield (t/ha) of sugarcane in India with other countries
VI	Export Import status of sugarcane produce
VII	Country wise export of sugar from India by ISEC during 2001 to 2010
VIII	Released and notified varieties of sugarcane
IX	Varieties resistant to stresses, suitability to biotic stresses
X	Sugarcane varieties tolerant to drought, water logging and salinity Developed in India
XI	State wise farmer's preferred / ruling varieties under cultivation
XII	Sugarcane, sugar and molasses production at a glance
XIII	Statement showing factories in operation, opening stocks, production, imports, consumption and exports of sugar (lakh tonnes) during last 10 years
XIV	Utilization of sugarcane for different purposes
XV	State wise utilization (%) of sugarcane for sugar production in major states
XVI	State wise cane crushed (in '000 tonnes) by sugar factories in India
XVII	Statement showing the minimum statutory price/ fair & remunerative price of sugarcane fixed by the government
XVIII	Per capita consumption of sugar, gur & khandsari
XIX	Temperature requirement for different growth stages of sugarcane
XX	Important growing / Planting and harvesting time of sugarcane in different States of India
XXI	Extent of losses due to different insect & pests in India
XXII	Management schedule for insect pests of sugarcane for Subtropical India
XXIII	Management schedule for insect pests of sugarcane for Tropical India
XXIV	Important Diseases of Sugarcane and Their Management
XXV	State wise number of sugar factories in operation in India
XXVI	State wise duration of crushing season for sugarcane in India
XXVII	State wise average sugar recovery percent cane in India
XXVIII	State wise & sector wise installed annual sugar production capacity and utilisation of capacity during last five years
XXIX	State wise sugar production ('000 ton) during last 10 years
XXX	Export- import of sugar on financial year basis from 2000-01 onwards
XXXI	State wise rate of cess/ purchase tax on sugarcane paid by sugar factories
XXXII	Progress report of sugarcane Development Scheme under Macro Management.
XXXIII	Statement showing the minimum statutory price/ fair & remunerative price of sugarcane fixed by the government

Chapter – 1: Crop description

1.1 Origin of the crop:

Cultivation of sugarcane in India dates back to the Vedic period. Barber (1931) was of the opinion that the thin Indian canes probably originated in the moist parts of north eastern Indian, from some plants closely related to *Saccharum spontaneum* (Kans). It belongs to family Gramineae (Poaceae), class monocotyledons and order glumaceae sub family panicoidae, tribe Andripogoneae and sub tribe saccharininea. The cultivated canes belong to two main groups: (a) thin, hardy north Indian types *S.barberi* and *S.Sinense* and (b) thick, juicy noble canes *Saccharum officinarum*. Highly prized cane is *S. officinarum*. It is probably evolved from *S. robustum* by introgression from other genera. It is agreed that the origin of S. officinarum is the Indo-Myanmar china border with New Guinea as the main centre of diversity. The process of nobilization in sugarcane is the modified back crossing of wild cane. *S. spontaneum* with *S. officinarum* and a repeated back crossings to the noble parent (*S. Officinarum*). It is stated that the cradle of cultivated sugarcane is the region where two wild species i.e., S*accharum spontaneum* and S*accharum robustum* are found. S*accharum robustum* is derived from natural crossing between S*accharum spontaneum* and *Miscanthis floridulus* and the origin is New Guinea.

1.2 Importance of Sugarcane:

Sugarcane is a most important cash crop of India. It involves less risk and farmers are assured up to some extent about return even in adverse condition. In agriculture sector, sugarcane shared 7% of the total value of agriculture output and occupied 2.6% of India's gross cropped area during 2006-07. Sugarcane provides raw material for the second largest agro-based industry after textile. About 527 working sugar factories were located in the country during 2010-11 with total crushing capacity of about 242 lakh tonnes. The sugar industry is an instrumental in generating the sizable employment in the rural sector directly and through its ancillary units. It is estimated that about 50 million farmers and their dependents are engaged in the cultivation of sugarcane and about 0.5 million skilled and unskilled workers are engaged in sugar factories and its allied industries. The sugar industry in India has been a focal point for socio-economic development in the rural areas by mobilizing rural resources, generating employment and enhancing farm income. Some of the sugar factories have also diversified into bye-products basis industries and have invested and put up distilleries, organic chemical plants, paper, ice board factories and cogeneration plant.

1.3 Scientific name of Sugarcane:

The genus *Saccharum* has six important species viz., with their common name

(i) *Saccharum officinarum,*
(ii) *S. Sinense*
(iii) *S.barberi*
(iv) *S.robustum*
(v) *S.spontanuem.*
(vi) *S.ilude*

The first three species are the cultivated species and remaining is wild ones. *S. officinarum* species is widely cultivated in India because of high sucrose content.

1.4 Morphology of Sugarcane:

Sugarcane is a tall perennial plant growing erect even up to 5 or 6 metres and produce multiple stems or culms each of which consist of a series of nodes separated by internodes. The plant is composed of four principal parts, root system, stalk, leaves and inflorescence. Following germination, the terminal vegetative bud of each shoot lays down a series of node. The root system is fibrous and consists of two types of roots, namely 'sett roots' and 'shoot roots'. When sugarcane sett is planted in the soil and covered with moist soil, the root primordia (translucent dots) situated at the base of every cane joint is activated and produces roots. These roots are known as 'sett roots' and are mostly temporary. After the emergence of the primary shoot from the bud, other roots are produced from lower rings of the lower nodes of the shoot. Later, this process occurs progressively in upper rings of the nodes near the soil surface. Those formed first go downwards, where as those formed near the soil. Surface roots grow in upper layer of soil for providing anchorage for the plant. These roots produced from shoot are known as 'shoot roots'. These are permanent roots and are thick, fleshy and white in colour. New roots are continually produced from tillers.

Sheath: Green with red blotches; moderate to heavy bloom; scarious border prominent; sheath splitting occasional Clasping; Spines present on the middle of the sheath; deciduous.

Blade Joint: Transverse Mark: Purplish green; medium: fair bloom. Ligule: Medium; Crescentiform; symmetrical; gradually tapering towards the edges. Ligular Process: Indicated on one side.

Inflorescence: The inflorescence of sugarcane generally called the 'arrow' is an open panicle. It is long (30 centimeter or more) and tapering. The arrangement of the spikelets is racemose, that is, the oldest flowers are at the bottom and the youngest at the top. The flowers open in succession over a number of days. Flowers have both male and female organs, but not all produce fertile pollen only few exception. Some of the varieties have fertile pollens but they are usually small and low vitality. Sugarcane usually flowers at the age of ten to twelve months, but some varieties in north India do not flower at all. Due to this fact cane has so long been propagated vegetatively by cuttings of sugarcane. Cane produced from seed is not so vigorous, but it is important for breeders.

Cane: Medium-thick; slightly staggered; slightly oval in cross section, internal tissue yellow with purple tinge: rind hard; pith present as small cavity. Yellowish green with purple tinge; turning dark green on exposure having red blotches; growth ring greenish yellow, root zone purplish yellow. Cylindrical with a tendency to become conoidal; splits present; ivory markings present; weather markings present; bud groove rather distinct; moderate bloom throughout. Wax Band: Medium; constricted often merges with the general bloom. Growth Ring: Medium; slightly swollen; occasionally the growth ring width and root zone width ale found to be almost equal. Root Zone: Narrow; slightly constricted; generally has two rows of staggered root eyes.

Node & bud: Nods are slightly depressed; leaf scar slightly inclined. Buds are medium, plumpy, ovate; occasionally hairs at the tip of the bud noticed; inserted at leaf scar.
Growth phases: Sugarcane is a C_4 plant having high efficiency in storing solar energy and most efficient converter of solar energy, thus having potential to produce huge amounts of biomass. Sugarcane has essentially four growth phases *viz*. Germination phase, Tillering phase, Grand growth phase, Maturity and ripening phase though it is difficult to recognize distinct duration of each.
(http://www.ikisan.com/crop%20specific/eng/links/ap_sugarcaneMorphology.shtml)

It is now widely accepted that India is the original home of *Saccharum species. Saccharum barberi* and *Polynesian* group of island especially New Guinea is the centre of origin of *S. officinarum.* Many of clones belonging to the *S. barberi* have been under cultivation in the sub tropical region of India since ages.

1.5 Nutritional value of Sugarcane:

The juice Sugarcane per serving (28.35 grams) contain Energy-111.13 kJ (26.56 kcal), Carbohydrates-27.51 g, Protein-0.27 g, Calcium11.23 mg (1%), Iron 0.37 mg (3%), Potassium 41.96 mg (1%), Sodium 17.01 mg (1%)
Source: Nutrient Information from ESHA Research

1.6 Important regions/ zones for sugarcane cultivation in India:

Broadly there are two distinct agro-climatic regions of sugarcane cultivation in India, viz., tropical and subtropical. However, five agro-climatic zones have been identified mainly for the purpose of varietal development. They are

i. North Western Zone
ii. North Central Zone
iii. North Eastern Zone,
iv. Peninsular Zone
v. Coastal Zone.

Tropical region shared about 45% and 55% of the total sugarcane area and production in the country, respectively along with the average productivity of 77 t/ha (2011-12). Sub-tropical region accounted for about 55% and 45% of total area and production of sugarcane with an average productivity about 63 t/ha (2011-12)

1.7 Tropical Sugarcane region

The tropical sugarcane region consists of sugarcane agro climatic zone 4 (Peninsular zone) and 5(Coastal zone) which includes the states of Maharashtra, Andhra Pradesh, Tamil Nadu, Karnataka, Gujarat, Madhya Pradesh, Goa, Pondicherry and Kerala. In the coastal areas of A.P. and Tamil Nadu there are extensive sugarcane areas with high sugarcane productivities. Floods, water logging and diseases such as red rot are the main problems. In the tropical region, sugarcane gets more or less ideal climatic conditions for its growth. It is cultivated with better package of practices and higher irrigation levels. The growing season is long with more equitable and favourable conditions without serious weather extremes. Being a tropical country, the agro-climatic conditions of tropical India favour higher sugarcane and sugar yields. The tropical region contributes about 55 per cent to the total cane production in the country. The productivities are higher. The average cane yields of the major states of the region (Maharashtra, Tamil Nadu, Karnataka, Andhra Pradesh and Gujarat) is around 80 tonnes per hectare. Maharashtra and the adjoining area of Karnataka, Gujarat and A.P. record higher sugar recoveries. Long hours of sunshine, cool nights with clear skies and the latitudinal position of this area are highly favourable for sugar accumulation. This is why average recoveries of Maharashtra and Gujarat are highest in the country. Moisture stress during the early part of the cane growths mostly during March to June, is an important problem.

1.8 Sub-tropical sugarcane region

Around 55 per cent of total cane area in the country is in the sub-tropics. U.P, Bihar, Haryana and Punjab comes under this region. Extremes of climate is the characteristic feature of this region. During April to June, the weather is very hot and dry. July to October is rainy season accounting for most of the rainfall from S-W monsoon rains. December and January are the very cold months temperature touching sub-zero levels in many places. November to March are cool months with clear sky. The N-W zone comprising the areas in Haryana, Punjab, Western Rajasthan and Western U.P. has very low temperature in December-January which often cause frost. During May and June, the temperatures are extremely high. Because of extremes of weather, the active sugarcane growth is restricted to 4-5 months only. In Eastern U.P, Bihar, and West Bengal, sugarcane suffers due to floods and water logging during monsoon months. The cane yields are lower in the subtropics due to short growing season, moisture stress, more pest and disease problem, floods and water logging and very poor ratoons. The average yield of the four major states (U.P, Bihar, Punjab and Haryana) is around 60 tonnes per hectare. However, there is considerable potential to be exploited.

1.9 Crop distribution:

Since 1930-31 the area of sugarcane in the country is increased by about 4.3 times and production by about 9.8 times because of the rise in productivity of sugarcane by about 2.28 times, however, during the last 10-15 years there is no much increase in productivity of sugarcane in the country.

1.10 Demand and supply scenario of sugarcane produce:

The main product of sugarcane is sugar which is produced in the sugar factories through various processes. The domestic demand of sugar is around 22-23 million tonnes annually where as the production of sugar in India is about 26 million tones annually. Up to the year 2011-12, Maharashtra is the largest producer of sugar contribute about 34% of sugar in the country followed by Uttar Pradesh.

Chapter – 2 : Comparative analysis

2.1 Area, production and yield of sugarcane in major growing states:

In Tropical zone Maharashtra is the major sugarcane growing state covering about 9.4 lakh ha area with production of 61.32 Million ton, whereas the productivity of Tamil Nadu is highest in tropical zones. Uttar Pradesh is the highest sugarcane producing State in sub tropical zone having area about 22.77 Lakh ha with the production of 135.64 Million Ton cane whereas Haryana has highest productivity of sugarcane in Sub tropical zone. Despite long tradition and large area in India, the average productivity of sugarcane is low with certain regions reporting yields as low as 40 t/ha only. The State wise and zone wise area, production and productivity of sugarcane during the last 5 years is given at ***Annexure- I (a,b,c)***. The production of sugar in India during last 10 year is given in **Annexure-II**. The area production and yield of sugarcane since 1930-31 is given at **Annexure-III**.

2.2 Area, production and yield of sugarcane in major growing countries:

Brazil is major producing country with an area about 90.77 lakh ha and production of about 717.46 Million ton followed by India. Amongst 10 major producing country Colombia has the highest yield of sugarcane due to the richest biodiversities in the world and has access to multiple climates. The area, production and yield of 10 major sugarcane producing country vis-e-vis world during the last 5 years is given at ***Annexure-IV***.

2.3 Gap of Yield with other countries:

There is a vast yield gap amongst the country with respect to India. Amongst the 10 major sugarcane growing countries the highest yield gap of 36.96 t/ha was recorded during 2009 between India and Colombia. The productivity of sugarcane in Philippines is also fairly high with respect to India. The yield gap of sugarcane in India with respect to 10 major sugarcane producing countries during the last 5 years is given at ***Annexure-V***.

2.4 Export import status of sugarcane produce:

Sugarcane as such is neither exported nor imported, however the sugar which is a main produce of sugarcane crop is exported and imported as per the Government policies. The export import of sugar during last 12 years is at ***Annexure-VI***. Country Wise Export (Tonnes) of sugar from India By ISEC during 2001 To 2010 is at ***Annexure-VII***.

Chapter 3: Varietal development

3.1 Important Sugarcane varieties released and notified are given at ***Annexure-VIII.*** Varieties resistant to stresses and suitability to biotic stresses are at ***Annexure-IX*** *and* tolerant to drought, water logging and salinity are at ***Annexure-X****.*

The ruling varieties of sugarcane in different States are given as under:

Andhra Pradesh:

Early varieties: Co.6907, 84A125, 81A99, 83A30, 85A261, 87A298, Co.8014, 86V96, 91V83.

Mid-late Varieties: COA7607, CO8021, COT.8201, Co7805, COV92102 (83V15), 83V288.

Late varieties: Co.7219, CoR8001, 87A380, Co7706

Bihar: Bo 99, CoP 9301, CoSe 98231, CoS 8436, Cos 95255, Bo 102, Bo 91, Bo 110, CoP 9206, CoSe 95422, CoSe 92423, UP 9530.

Gujarat: Co 86002, Co 86032, CoSi 95071, Co 86249, CoN 05072.

Haryana: CoJ 64, CoS 8436, CoS 88230, CoS 767.

Karnataka: Co 94012, CoC 671, Co 92020, Co 8014, Co 86032, Co 62175, Co 8371, Co 740, Co 8011.

Maharashtra: CoC 671, Co 86032, Co 8011, Co 94012, CoM 265, Co 92005.

Odisha: Co 62175, CoA 89085, Co 87A298, Co86V96

Punjab: CoJ 85, CoJ 88, CoS8436, CoH 119, Co89003.

Tamil Nadu: Co 94012, Co 94010, CoC 24.

Uttar Pradesh: CoS 8436, Coj 64, CoS88230, CoS 98231, CoS 767, CoS 8432, CoPt 90223, CoS 92423, CoS97264, CoLk 8102.

Uttrakhand: CoS 8436, CoS 88230, Cos 767, CoS 97264, CoSe 92423.

State wise farmer's preferred / ruling varieties under cultivation separately for less than ten years old and more than 10 years old – with reasons towards their preference by farmers is given at ***Annexure-XI***

3.2 State wise yield potential recorded under FLDs (AICRP Projects) vis-à-vis National/ State average yield and yield gap analysis.

The state wise yield potential recorded under FLDs vis-à-vis state average yield and gaps in yield are given below:

The State wise/technology wise yield gap during the year 2009-10

Sl.No.	State	Technology demonstrated	Potential yield (FLD) (t/ha)	State yield (t/ha)	State Yield gap (t/ha)	National Yield gap* (t/ha)
A.	**Tropical region**					
1	**Andhra Pradesh (2009-10)**	Ratoon Management	84.00	74.11	9.89	13.95
2	**Gujarat (2009-10)**	Integrated disease and pest management	112.00	80.52	31.48	41.95
		Drip fertigation	126.50		45.98	56.45
		Intercropping with oilseeds (S-82.00t/ha & G-980 Kg/ha.)				
		Ratoon Management	112.19	84.87	12.19	42.14
3	**Maharashtra (2009-10)**	Weed Management	130.90		46.03	60.85
		Ratoon management	170.46		85.59	100.41
		Full package of practice	272.00		187.13	201.95
4	**Karnataka (2008-09)**	Integrated nutrient management	149.10	88.65	60.45	79.05
		Relay cropping of sugarcane with tobacco S-110.86 & T-13.16				
		Dual row planting and drip irrigation	128.63		39.98	58.58
		Ratoon management	112.68		24.03	42.63
5	**Tamil Nadu (2008-09)**	Wider row planting	132.50	108.15	24.35	62.45
		Paired row with drip fertigation	178.00		69.85	107.95
		Performance of bio-fertilizers	146.50		38.35	76.45
6	**Kerala (2009-10)**	Ratoon management	88.07	95.00	-6.93	18.02
		Irrigation Management	108.64		13.64	38.59

B.	**Sub-tropical region**					
1	**Uttarakhand (2009-10)**	Inter-cropping with Mung bean with sugarcane	75.70	60.83	14.87	5.55
2	**U.P. (2009-10)**	Recommended package of practice	123.50	59.25	64.25	53.45
		Trash mulching	85.50		26.25	15.45
3	**Punjab (2009-10)**	Integrated weed management	91.80	61.67	30.13	21.75
		Management of late planted sugarcane	83.50		21.83	13.45
		Integrated disease and pest management	75.00		13.33	4.95
		Integrated nutrient management	97.00		35.33	26.95
		Inter-cropping with summer mung	101.30		39.63	31.25

* National yield = 70.05 t/ha.

Source: Progress report during AICRP Meeting 2009-10.

3.3 In India 2011-12 about 263.43 Lakh ton sugar and 118.24 lakh ton molasses were produced by 529 sugar factories with average sugar recovery of 10.25%. The area, production, yield, number of factories, sugar and molasses production and sugar recovery is given at ***Annexure-XII.*** In India out of 263.43 lakh ton sugar produced during 2011-12, 220 lakh ton is consumed domestically and 33.9 lakh ton exported. The details of number of factories in operation, opening stocks, production, imports, consumption and exports of sugar (lakh tonnes) during last 10 years is given at ***Annexure XIII.*** *On an average 72% of cane used for the production of white sugarc, 16% for gur and Khandsari and rest of about 12% is used for seed and feed purposes. The details of* utilization of sugarcane for different purposes is given at ***Annexure XIV.*** Amongst the Indian States, Maharashtra recorded (94.14%) utilization of sugarcane for the production of sugar and Uttrakhand recorded least (30.99%). The details of state wise utilization (%) of sugarcane for sugar production in major states is given at ***Annexure XV****.* In India, Maharashtra State crushed maximum sugarcane followed by Uttar Pradesh. The state wise cane crushed (in '000 tonnes) by sugar factories in India is given at ***Annexure XVI.***

3.4 For the year 2012-13 the Government of India has fixed fair and remunerated price for sugarcane as Rs. 170/qntl linked with 9.5% sugar recovery and premium of Rs. 1.79/qntl for every 0.1 % increase in sugar recovery. The details of price trend fixed by the government is given at ***Annexure XVII***. On an average an Indian consumes about 23 Kg sugar, fur and khansari in a year. The per capita consumption of sugar, gur & khandsari is given at ***Annexure XVIII.***

Chapter 4: Climatic requirement

4.1 Temperature for different critical stages of sugarcane:

The critical stages of germination are tillering, early growth, active growth and elongation. Optimum temperature for sprouting (germination) of stem cuttings is 32° to 38°C. It slows down below 25°C, reaches plateau between 30°-34°C, reduces above 35°C and practically stops when the temperature is above 38°C. Temperatures above 38°C reduce the rate of photosynthesis and increase respiration. For ripening, however, relatively low temperatures in the range of 12° to 14° are desirable. The temperature requirement for different growth stages of sugarcane is given at ***Annexure-XIX***

4.2 Reduction in yield of sugarcane due to rise in temperature:

The sugarcane productivity and juice quality are profoundly influenced by weather conditions prevailing during the various crop-growth sub-periods. Sugar recovery is highest when the weather is dry with low humidity; bright sunshine hours, cooler nights with wide diurnal variations and very little rainfall during ripening period. These conditions favour high sugar accumulation. The climatic conditions like very high temperature or very low temperature deteriorate the juice quality and thus affecting the sugar quality. Favourable climate like warm and humid climate favour the insect pests and diseases, which cause much damage to the quality and yield of its juice and finally sucrose contents. Short days of winter reduce tillering, produce thinner long plants, with less sucrose contents. Heavy rains create water logging conditions *(www.sugarcanecrops.com).*

4.3 Recommendation for cultivation of crop in view of climate change:

Several geopolitical factors, aggravated by worries of global warming, have been fueling the search for and production of renewable energy worldwide for the past few years. Such demand for renewable energy is likely to benefit the sugarcane ethanol industry, not only because sugarcane ethanol has a positive energetic balance and relatively low production costs, but also because ethanol has been successfully produced and used as biofuel in the country since the 1970s. However, environmental and social impacts associated with ethanol production can become important obstacles to sustainable biofuel production worldwide. Atmospheric pollution from burning of sugarcane for harvesting, degradation of soils and aquatic systems, and the exploitation of cane cutters are among the issues that deserve immediate attention. The expansion of sugarcane crops to the areas presently cultivated for soybeans also represent an environmental threat, because it may increase deforestation pressure from soybean crops. Recommendations include proper planning and environmental risk assessments for the expansion of sugarcane to new regions, improvement of land use practices to reduce soil erosion and nitrogen pollution, proper protection of streams and riparian ecosystems, banning of sugarcane burning practices, and fair working conditions for sugarcane cutters.

Read More: http://www.esajournals.org/doi/abs/10.1890/07-1813.1

Chapter-5: Genetic Potentiality advancement

5.1 Genetic break through for yield improvement from ICAR / SAUs/ International organizations:

There is no commercially grown transgenic sugarcane. Research and development of transgenic sugarcane has been identified in: Australia, Argentina, Brazil, Cuba, Egypt, India, Indonesia, Mauritius, Myanmar, South Africa, USA, Venezuela

Brazil: First Transgenic Sugarcane:

Brazil, the top producer of ethanol derived from sugarcane has developed its first transgenic sugarcane that contains the gene for drought tolerance. The research in Brasilia at the Empresa Brasileira de Pesquisa Agropecuaria (EMBRAPA) on the introduction of drought tolerant gene DREB2A into sugarcane was started in 2008. Through biolistic transformation method, transgenic sugarcane plants were developed and evaluated and selected in a greenhouse. The transgenic plants will be evaluated for their tolerance to drought by May 2012. Selection of the best performance both in the agronomic and desired characteristics will follow the evaluation process set by the National Technical Committee (CTNBio). This is the first transgenic sugarcane developed which will open possibilities for the introduction of other traits into sugarcane beneficial to farmers, consumers and the industry.

Source: Crop Biotech Update, 27th May 2011

5.2 Status of Transgenic Sugarcane in India

The Sugarcane Breeding Institute, Coimbatore has developed transgenic sugarcane incorporating gene constructs. The molecular marker technologies offer more precision to sugarcane breeding, transgenic technology gives a new dimension to the varietal development programmes. Scientists at the Coimbatore-based premier Sugarcane Breeding Institute (SBI) have developed transgenic varieties, which are resistant to the dreaded red-rot disease and borer pests. They identified the genes resistant to the red-rot and pests from indigenous sugarcane species, particularly from the wild, in collaboration with the New Delhi-based National Research for Plant Biotechnology (NRPB). The scientists have developed the transgenic varieties by bombarding the genes with a particle gun into the subject sugarcane plant known for very high sucrose yield such as COC-671 and screened them for red-rot disease resistance.

5.3 Advance tools to be applied if any like transgenic, genomics etc.:

SBI has used the molecular technologies in introgression of wild species and developed sugarcane transgenics that cane agriculture faced serious challenges in terms of sustainability. High cost of production, depleting natural resources, climate change, non-availability of labour, emerging new pests and diseases have impacted cane productivity and sustainability. Further studies conducted in India, Mauritius, South

Africa and Trinidad showed a 30 per cent and more loss in productivity for every two degree centigrade increase in temperature. Ethanol production: While sustainability has been a cause for concern in cane agriculture, the demand for sugar in the country by 2030 has been estimated at 36 million tonnes for which cane production should be 500 million tonnes. This is 40 per cent higher than the current production. Besides this, sugarcane itself is emerging as an important energy crop contributing to cogeneration and ethanol production. The current level of ethanol production is barely adequate to meet 10 per cent blending and the demand is expected to soar in the coming years. Cane production needs to be enhanced to meet the ethanol requirement as well. The cogeneration potential in the country has been estimated at 5,500 MW, while the present installed capacity is just about half at 2,500 MW. Transgenic sugarcane can increase yields, reduce production costs, improve sugar quality and reduce the environmental impact of sugarcane cultivation. But, Each transgenic event must be analysed separately, because the impact (benefit and risk) of each trait will be different. Strong and reliable regulatory agencies are needed.

Chapter – 6: Seed Scenario

6.1 The normal practice in sugarcane growing States of country is to use commercial crop of sugarcane for seed purposes. Sugarcane is vegetatively propagated and required huge quantity of seed. The accounting of different classes of sugarcane seed i.e. breeder, foundation and certified are not being maintained by the different sugarcane growing States therefore the exact quantum of certified seed distributed by different agencies in major sugarcane growing states could not be assessed. The important cane seed production advanced technologies are available as under:

6.2 Tissue culture

The tissue culture technique in sugarcane can be used for rapid multiplication of newly developed high yielding, high sugar, disease resistant varieties and rejuvenation of outstanding varieties under cultivation. The vegetative propagation of sugarcane through seed cane cuttings is cumbersome requiring larger quantities of vegetative seed material and seed multiplication rate is too low, needing about ten years for a new variety to be released for cultivation and cover larger areas in the field subsequently. With tissue culture technique, it is possible to release a variety within five years and propagate it quickly in the field. The productivity of the important varieties can also be maintained and improved for their longer life span in the field. The micro propagation technique used in this technology has the following advantages:

- Production of true to type plantlets
- rapid multiplication
- independent of seasonal constraints
- maintaining and improving the productivity of outstanding varieties in the field
- production of disease free planting material from apical meristem

The tissue culture technique has the following important components:

i. Taking of spindle explants/Apical meristem from moist hot air treated seed cane crop of six to eight months age.
ii. Media preparation
iii. Inoculation
iv. Multiplication of cultures
v. In vitro rooting
vi. Pre hardening
vii. Transfer of plantlet from laboratory to Polybags
viii. Hardening
ix. Transplanting in the field

6.3 Polythene Bag Technology

In this technique, fertile soil is filled in one kg. Polythene bags, small holes made in the bottom portion of polythene bag for aeration, single bud cuttings made from top one third portion of seed cane stalk, planted horizontally with bud position upward in top one fourth soil portion and covered with thin soil layer followed by applying water with a watering cane. Alternatively, polythene bags are placed in a ploughed up field plot and irrigation water applied through surface irrigation. In this technique, the germination is almost 95% while in direct vegetative seed cane planting in the field, the germination hardly 40-50%. These pre germinated shoots in polybags are ready for transplanting in the field 30 days after planting by which time they attain 4-5 leaf stage. The poly cover is removed with a blade at the time of transplanting and the plantlets along with the earth ball placed in a mini pit made in the furrow and filled with soil and irrigation water applied. With this technique, only 20 qtl. seed is sufficient to transplant one hectare area while in normal direct sett planting about 75 qtl. seed is required. Therefore, there is significant saving of seed cane with this technique. This technique has particularly been found very useful for quick multiplication of seed of new promising varieties, replanting of ratoons and filling of gaps. It is also useful for late planting of sugarcane after wheat harvesting in April-May.

6.4 Raising of seedlings through bud chip:

Production of sugarcane seed with high quality utilizing low cost technologies is the need of the hour. Transplanting of settlings raised in polybags offers great advantages **in** sugarcane seed production. Nursery raised from sugarcane bud chips and planting them in main field was found to be more economical than traditional methods. At Sugarcane Breeding Institute, an improved manually operated Bud Chipping Machine was fabricated. A high tension steel blade is used to scoop the bud as chips from a horizontally fed cane. Techniques for making bud chips, their storability, transport, raising the seedlings in polythene bag as well as their survival and establishment on a suitable medium were standardized. Bud chips could be taken using Bud Chipping Machine, treated with pesticide solution and transported to the destination for seed production.

6.5 Spaced transplanting technology:

The technology as described by Srivastava et. al. (1981) consist of transplanting of nursery raised settlings. Settling are raised by planting of single bud set in nursery about one month before transplanting in the main field. About 2 tonnes of cane seed are required to obtain settlings for transplanting in one hectare of field. Apart from above conventional method of cane seed production is very popular through out India.

Chapter 7: Improved crop production practices

7.1 Major crop sequences/ rotations followed in various states and suggestion crop sequences by SAUs/ ICAR

Table: Prominent sugarcane based cropping system in tropical and subtropical regions of the country recommended.

Sub tropical	Tropical region
Paddy- Autumn Sugarcane-ratoon-wheat	Bajra-Sugarcane(pre-seasonal)-Ratoon-wheat
Greengram- Autumn Sugarcane-ratoon-wheat	Paddy-Sugarcane-Ratoon- Finger millet
Maiz- Autumn Sugarcane-ratoon-wheat	Paddy-Sugarcane-Ratoon- Wheat
Kharif Crops-Potato-Spring Sugarcane-ratoon-Wheat	Paddy-Sugarcane-Ratoon- gingelly
Kharif Crops-Mustard-Spring Sugarcane-ratoon-Wheat	Paddy-Sugarcane-Ratoon- urd.
Kharif Crops-Pea/Coriander-Spring Sugarcane-ratoon-Wheat	Cotton-Sugarcane-Ratoon–wheat
Kharif Crops-Wheat-late Planted Sugarcane-ratoon-Wheat	Sugarcane-Ratoon-Kharif rice-Winter rice.

7.2 State and season wise time of sowing and harvesting:

Sugarcane take generally one year to mature in sub tropical states (U.P., Punjab, Haryana, Bihar etc.) called “Eksali” however in some tropical states it matures in 18 months (Andhra Pradesh, Karnataka, Maharashtra etc.) called “Adsali”. In India, the planting seasons of sugarcane in different States is given at ***Annexure-XX.***

7.3 Planting of sugarcane crop:

(i) Different season and method of planting: Sugarcane can be planted as per the recommendation for the region i.e. Autumn Planting (15 Sept. to Oct.) and Spring Planting (Feb. to March). Improved method of planting should be adopted like, deep furrow, trench methods, ring pit method and paired row method instead of furrow system.

(ii) Land preparation by use of different technologies like RCT:

7.4 Resource Conservation Technology (RCT) in Sugarcane:

Application of nitrogen fixing (*Azospirillum* and *Gluconacetobacter*) and phosphate solubilizing (Phosphobacteria) bio-fertilizers were found to reduce the requirement of chemical fertilizers to the extent of 25%. Reduction in the dose of chemical fertilizers reduces soil degradation (Source: http://www.sugarcane.res.in). Trash mulching of dry leaves, drip irrigation for water saving and mechanization through Ratoon management device (RMD), sugarcane cutter planter, trench opener, power weeder etc. are successfully using for saving for man power as well as time.

7.5 Seeding technologies –seed rate, distance, depth, plant population

Seed rate:

Seed rate in sugarcane varies from region to region. Generally higher seed rate are used in north western India (Punjab, Haryana and Rajasthan) because of the lower germination percent and also adverse climatic condition (very hot weather with desiccating winds) during tillering phase. A northern region seed rate generally varies from 40,000 to 60,000 three budded setts per hectares while in southern region it range between 25,000 to 40,000 three budded setts.

Row spacing:

Effect of row spacing from 45 to 120 cm has been tried on growth, yield and quality of sugarcane. Optimum inter rows spacing range between 60-100 cm under different situation and location.

Optimum seed rate and row spacing for sugarcane in different states in India

S.No.	State	Seed rate (3 buded sett (000/ha)	Row spacing (cm)
Sub-tropical region			
1	Uttar Pradesh		
	Timely planting	35	90
	Late Planting	56	60
	Bihar	37	75-90
	Punjab	50	60-75
	Haryana	75 (2 budded)	60-75
	Rajasthan	40-45	75-90
	Madhya Pradesh	25-30	90
	Assam	37-42	90
	West Bengal	25-30	90
Tropical			
	Maharashtra	30	90-100
	Andhra Pradesh	30	80-90
	Karnataka	25-30	90
	Gujrat	25-30	90
	Tamilnadu	42	80
	Orissa	37-40	90
	Kerala	35-40	90

Source: Sugarcane Production technology in India by Dr. R. S. Verma

Depth:

About 80% of the sugarcane roots go up to a depth 60 cm. Hence deep ploughing of sugarcane fields is necessary. Initially one or two deep ploughings with tractor drawn disc plough or mould board plough or animal drawn mould board plough have to be done at least to a depth of 30 cm. This has to be followed by ploughing with other light tillage implements.

7.6 Fertilizer management – recommended dose for different ecologies, micro nutrients, organic manure, application method

An average crop of sugarcane yielding 100 t/ha removes 208kg of N, 53kg of P, 280kg of K, 30 kg of Sulphur, 3.4kg of iron, 1.2 kg of manganese, 0.6 kg of copper respectively from the soil. Hence, soil has to be replenished to maintain the productivity of sugarcane with the said quantities of nutrients.

Micro nutrient management

If the soil test value is below the critical value, apply sulphate form of Zn, Cu, Fe and Mn through soil application and foliar spray (The total concentration of salt should be 0.5% for young crop and 2.5% for a grown up crop).

Nutrient concentration in soil (for sugarcane)

Nutrient	Critical levels
(i) Fe (non-calcareous soil)	4.2 ppm
(ii) Fe (calcareous soil)	6.3 ppm
(iii) Zn (Loamy soils)	1.2 ppm
(iv) Zn (Clay soils)	2.0 ppm
(v) Mn	2.0 ppm
(vi) Cu	1.2 ppm
(vii) Hot water soluble-B	0.44 ppm

Organic matter helps in better soil structure formation and provides the most favourable air and water regimes. It is the source of plant nutrients including micro-nutrients, which are liberated in available form during mineralization. It increases the water holding capacity, buffer and exchange capacity and microbial activity of the soils. Hence soil organic matter is considered as an elixir of soil productivity. The recommended dose of bio-fertilizers for sugarcane crop is 10-12 kg/ha Acetobector, Azotobector, *Azospirillum* (or *Gluconacetobacter*) and PSB are the major bio fertilizers which are being used in Sugarcane crop.

7.7 Water management: application and conservation methods, their water use efficiency, water requirement of crops, critical stages for irrigation and probable losses if not applied:

In tropical area, irrigations are to be given once in 7 days during germination phase (1 –35 days after planting), once in 10 days during tillering phase (36 – 100 days after planting), again once in 7 days during grand growth phase (101 – 270 days after planting) and once in 15 days during maturity phase (271 days after planting up to harvest) adjusting it to the rain fall pattern of the area. About 30 to 40 irrigations are needed. Sugarcane is a high water requirement crop. About 250 tonnes of water is needed to produce one tonne of sugarcane. Methods like alternate furrow irrigation, drip irrigation and trash mulching could be of use to economize irrigation water during water scarcity periods. Foliar spraying of a solution containing 2.5% urea and 2.5% muriate of potash 3 or 4 times at fortnightly intervals during drought periods would help to reduce the impact of drought on the crop (*Source:* http://sugarcane.res.in)

7.8 Water requirement and applying irrigation at critical stages of growth:

As mentioned earlier, critical stages are those during which sugarcane is affected severely due to water stress and the loss cannot be restituted by adequate water supply at later stages. These stages are: sprouting (germination), formative stage or tillering, ripening and initiation of sprouting in ratoons. In case of limited water availability, one may sustain sugarcane productivity by irrigating at critical stages of growth. (Sustaining sugarcane productivity under depleting water resources: (Current Sc 748 IENCE, Vol. 101, No. 6, 25 September 2011 by Ashok K. Shrivastava, Arun K. Srivastava and Sushil Solomon are in the Indian Institute of Sugarcane Research, Lucknow 226 002, India.)

Water requirement (WR) in various sugarcane- growing states of India

State	WR (ha-cm)
Tropical India	
Andhra Pradesh	160–170
Tamil Nadu	180
Karnataka	200–240
Maharashtra	
(i) Plant cane (seasonal)	250
(ii) Plant cane (pre-seasonal)	300
(iii) Plant cane (Adsali)	350
(iv) Ratoon	300
Madhya Pradesh	270
Subtropical India	
Bihar	140
Uttar Pradesh	160–180
Punjab	170–180

Effect of irrigation methods on sugarcane yield, water-saving and water-use efficiency:

Agronomic measure	No. of demonstrations	Cane yield (t ha–1)			Water applied (cm)			IWUE (kg ha–1 cm)		
		D	FP	Increase (%)	D	FP	Saving (%)	D	FP	Increase (%)
TM	10	75.9	54.9	38.2	42.5	59.6	28.7	1786.4	921.5	93.9
SF	14	82.8	63.7	30.0	48.7	63.9	31.2	1700.2	996.2	90.6
ICGS	9	82.8	63.4	30.6	46.0	55.3	16.81	1800.0	1146.5	57.0

D, Demonstration; FP, Farmer's practice; IWUE, Irrigation water use efficiency; TM, Trash mulching; SF, Skip-furrow method of irrigation; ICGS, Irrigation at critical growth stages

Source: (Sustaining sugarcane productivity under depleting water resources: Current Sc 748 IENCE, Vol. 101, No. 6, 25 September 2011 by Ashok K. Shrivastava, Arun K.

Srivastava and Sushil Solomon are in the Indian Institute of Sugarcane Research, Lucknow 226 002, India.)

7.9 Weed Management – important weed flora, herbicides recommended with dose application time and different methods (mechanical, biological etc.)

In sugarcane, weeds have been estimated to cause 12 to 72 % reduction in cane yield depending upon the severity of infestation. The nature of weed problem in sugarcane cultivation is quite different from other field crops because of the following reasons:

- Sugarcane is planted with relatively wider row spacing.
- The sugarcane growth is very slow in the initial stages. It takes about 30 – 45 days to complete germination and another 60-75 days for developing full canopy cover.
- The crop is grown under abundant water and nutrient supply conditions.
- In ratoon crop very little preparatory tillage is taken up hence weeds that have established in the plant crop tend to flourish well.

The major weeds are Sedges- Cyprus rotundus; Grasses-Cynodon dactylon, Sorghum helepense, Panicum spp, Dactylocternium aegyptium, Broad leaved weeds – Chenapodium album, Convolvulus arvensis L., Amaranthus viridis L., Portulaca oleraceae L., Commelina bengalensis L., Trianthema portulacastrum L. etc.

Integrated weed management

Complete weed control cannot be achieved by using any one method. To have more dependable, economical and desirable weed control without environmental problems, it is advisable to have a proper combination of agronomical, cropping, rotational and biological methods with supplemental use of herbicides. By continuous use of herbicides, in addition to the residual problem there is every possibility of a major weed becoming a problematic one. Several of the effective herbicides are either non-available or prohibitively priced and therefore their use is restricted. An integrated approach wherein only supplemental use of herbicide is resorted to is highly desirable. Integrated weed management involves use of all feasible control measures such as mechanical, cultural, cropping, biological and chemical, in proper combination at different stages of weed and crop growth to get the most practical and economical results.

Mechanical control

- In integrated weed management, mechanical and cultural methods assume great importance. The most important aspect of weed management is checking the weed seed dispersal. This calls for destroying weeds before they flower and set seeds. Mechanical destruction is an effective measure.
- Keeping the irrigation channels clean helps in checking transport and dispersal of weed seeds. A thorough tillage, besides providing a good seedbed to the sugarcane crop, also destroys weeds.

Agronomic practices

- Growing the crop with proper agronomic practices will make the crop more competitive. Maintaining adequate crop population and application of optimum doses of nutrients are important.
- In ratoon sugarcanes, gaps favour weed growth. An adequate ratoon population must be achieved by stubble shaving, off barring and gap filling.

Pre-emergence herbicide

- Use of a pre-emergence herbicide is essential to control the weeds during the germination phase of the crop.
- Post-Emergence Management of Parasitic weed *Striga asiatica* in Sugarcane is new practice developed by TNAU in which pre-emergence application of atrazine 1.0kg/ha on third day after planting + hand weeding on 45 DAP with an earthing up on 60 DAP combined with post-emergence spraying of 2,4-D sodium salt 5g/litre (0.5%) + urea 20g / litre (2%) on 90 DAP for complete control is adopted. So that complete control of parasitic weed Striga Asiatica is ensured. Seed production by the weed is also avoided. This is an environmentally safe technology.

Hoeing earthing up and hand weeding

- Hoeing the inter rows using a ridger would help control weeds besides being able to earth up the cane rows. Hand weeding of the weeds in the cane rows is still necessary.
- Earthing-up operation is also known as "hilling-up". This operation is carried out in two or three stages. The first earthing-up operation is known partial earthing-up and the second/third operation is known as "full earthing-up". The partial earthing-up is done at 45 days after planting. While doing partial earthing-up, the furrow in which the cane row is present gets partially filled-up. Full earthing-up is done after 120 days after planting coinciding with the peak tiller population stage. During full earthing-up the soil from the ridge in between is fully removed and placed near the cane on either side. This operation converts the furrows into ridges and ridges into furrows. This operation could be done either manually or by using a bullock-drawn/tractor drawn furrower depending upon the spacing adopted. One more earthing-up after cane population is stabilized at 180 DAP may be helpful in preventing lodging and water shoots formation. It also improves aeration and helps to control weeds.

Irrigation

- Avoiding excess irrigation reduces the build up of weeds and damage to the soil.

Mulching

- Sugarcane trash can be profitably utilised for moisture conservation, weed control and to enrich organic matter status. Trash mulch reduces weed growth.
- Artificial mulches have been used for intercepting sun-light reaching the soil surface and thus starve out weed seedling as they emerge. Uniform spread of 5 to 10 cm

thick trash blanket in-between the rows, as also in the inter-plant spaces help in suppressing weeds. The trash keeps all weeds suppressed and the fields remain almost free from weeds. It is necessary to give a thorough hoeing and weeding in the field before spreading the trash, particularly if the field has been weedy. In places, where obnoxious grass weeds are in abundance the trash cover has to be a little thicker, say 10 cm to 15 cm thick. This is indeed a very potent and useful way of suppressing weeds in sugarcane fields. Beside weed control, trash mulching saves water, labour and costs, and gives higher yields of cane. Trash-mulched plots are remarkably easy on irrigation. Bagasse, paddy husk, hay, straw, etc. can also be used as mulching material. Spreading the black polythene film (Solarization) in interspaces suppresses the growth of weeds and conserves soil moisture.Intercropping

- In sugarcane, black gram, green gram, soybean, sunflower, groundnut etc. can be grown as intercrops, and they help in smothering weeds.

Crop rotation

- Rotating sugarcane with other crop reduces the incidence of certain weed species, for example, puddling and growing paddy helps in reducing the incidence of several grass and broad leaved weeds.

Weed Management in Pure Crop of Sugarcane

i. Spray Atrazine 2 kg or Oxyflurofen 750 ml/ha mixed in 500 ltr. of water as pre emergence herbicide on the 3rd day of planting, using deflector or fan type nozzle.
ii. If pre-emergence spray is not carried out, go in for post-emergence spray of Grammaxone 2.5 litre + 2,4-D sodium salt 2.5 kg/ha in 500 litre of water on 21st day of planting.
iii. If the parasitic weed striga is a problem, post-emergence application of 2,4-D sodium salt @ 1.25 kg/ha in 500 litre of water/ha may be done. 2, 4-D spraying should be avoided when neighbouring crop is cotton or bhendi.
iv. Apply 20% urea also for the control of striga as direct spray.
v. Pre- plant application of glyphosate at 2.0 kg ha-1 along with 2% ammonium sulphate at 21 days before planting of sugarcane followed by post emergence direct spraying of glyphosate at 2.0 kg ha-1 along with 2% ammonium sulphate with a special hood on 30 DAP suppressed the nut sedges *(Cyperus rotundas*) and provided weed free environment.
vi. If herbicide is not applied work the junior-hoe along the ridges on 25, 55 and 85 days after planting for removal of weeds and proper stirring.
vii. Remove the weeds along the furrows with hand hoe. Otherwise operate power tiller fitted with tynes for intercultivation.

Weed management in Sugarcane intercropping system

Premergence application of Thiobencarb @ 1.25 kg ai/ha under intercropping system in Sugarcane with Soybean, blackgram or groundnut gives effective weed control.

7.10 Plant protection – important insects pests and diseases and their effect on yield loss, chemical, biological and cultural control measures:

Sugarcane is liable to be attacked by a number of insect pests and diseases. According to an estimate, sugarcane production declines by 20.0 and 19.0 % by insect pests and diseases respectively. To increase the crop productivity, management of insect-pest and diseases is of great significance. Due to diversity in agro-ecological conditions the importance of insect pests and disease varies and therefore, management strategy should be adopted accordingly.

Sugarcane is infested by about 288 insects of which nearly two dozen causes heavy losses to the quality as well as quantity of the crop. The scenario of insect pests and diseases varies in sub-tropical and tropical belt of sugarcane. Top borer and stalk borer are found pre-dominantly in sub-tropical areas whereas internodes borer and early shoot borer and among disease rust & eye spot are prevalent in tropical region. The extent of losses due to different insect & pests in India. Management schedule for insect pests of sugarcane for subtropical India is given at ***Annexure-XXI.*** Management schedule for insect pests like Borers (stem, shoot and internode), scale insect, termites, shoot borer, root borer, thrips, White grubs, woolly aphid, White fly, snail, rodents etc. of sugarcane for Subtropical & Tropical India is given at ***Annexure-XXII & Annexure XXIII.*** Important diseases like Red rot, smut, wilt and grassy shoot and mosaic etc. of sugarcane and their management is given at ***Annexure-XXIV.***

7.11 Harvesting:

Harvesting and collection of cane should be either mechanical or manual. It has been found that when cane is harvested and gathered mechanically, by combined harvester, or manually cut and collected and then grab loaded into large trucks/ tractor trolley.

Time of harvest: As far as possible harvesting should be carried out avoiding extremes of weather and within a short space of time, in order to obtain an even stand of ratoon cane. When crop is harvested in hot dry months, stubble and the soil tend to dry up quickly. Sprouting would be low due to drying up of the stubble and buds. Some times, immediately after ratooning, there may be a short wet spell followed by a long dry period, which will again lead to ratoon failure, as the shoot mortality will be high in the long dry period. In sub tropical of India, it has been shown that spring harvested plant crop-would result in a better ratoon then that obtained by harvesting in the autumn period. It has been observed that ratooning during winter months would lead to failure in axillary bud sprouting causing poor stands and much yield loss (Agrawal and Srivastava, 1983; Misra and Mathur, 1986; Yadav, 1993). In subtropical India, rationing of cane from mid January through February is preferred in order to obtain a high shoot population (Srivatava et al. 1985). Mishra and Mathur, 1991 have noted the influence of harvest time and fertility level of soils on multiple rationing of sugarcane in India. Although the optimum temperature range for the growth and sprouting of sugarcane is 26-30^{o}C and subtropical areas, because of the poor sprouting depth occurs in low temperature periods, most of the agro-technologies is concerned the establishment of high population density of millable stocks at maturity in the plant crop.

Harvesting technique: Harvesting of vegetative stocks, automatically initiates the regeneration of ratoon crop, with the removal of apical dominance, the stable is piece is now free to sprout its axillary buds from top downwards. In the manual system of harvesting using a straight blade knife, generally 5-10 cm piece of stable is left above the ground. The sugarcane crop can be cut at the ground level, then axillary buds which are below the ground level are forced to sprout and ratooning occurs with a well established root system which will be able to withstand wind as well exploit the total ground area. It has been found that knife with a curved blade in much superior to straight knife in harvesting sugarcane sticks in flush with the ground (Meemeduma, 1983).

7.12 Use and recommendation of farm implements and machines used for different operations:

Mechanization in Sugarcane Cultivation

In the competitive involvement, which is fast enveloping all the operations, the effectiveness of human resources will be the cutting edge for sustaining the productivity. Mechanization in Indian agriculture has proceeded along with the time tested two pronged approach based on improved equipments and enhance power supply. However, compared to the mechanization of western agriculture which was motivated by the need to substitute human labour and draught animal with mechanical prime movers, to maintain the socially desirable mix of human labours, draught animal power and mechanical power. Indian agriculture, powered by about 185 million agriculture workers, 73 million draught animals and 3 million tractors, reflect the unique Indian approach to mechanization. The analysis of impact of mechanization has shown that the emphasis on timeliness, precision and general improvement in the quality of work, mechanization has undoubtedly contributed to increase the yield, cropping intensity and total production and cost effectively. These machines have been developed for different farm operations such as tillage, seed bed preparation, sowing and planting, weeding and inter culture operations, irrigations, plant protection and harvesting etc. Sugarcane harvester is mainly using for harvesting of sugarcane in tropical states.

Machinery adopted for mechanical sugarcane cultivation in India

Animal drawn sugarcane planter: This equipment plants pre-cut sugarcane sett that are manually metered. It has a 3 wheeled fore-carriage and a trailed implement attached behind it.

Tractor drawn mustard Drill attachment to semi automatic sugarcane planter: It is an attachment to tractor mounted IISR semi-automatic sugarcane planter for sowing mustard and sugarcane, developed by IISR, Lucknow.

Power operated sugarcane sett cutting machine: It is a power-operated machine for cutting sugarcane setts for planting developed at IISR, Lucknow.

Tractor drawn semi automatic sugarcane planter: It is a tractor mounted implement suitable for planting sugarcane setts and application of granular fertilizer developed by IISR Lucknow.

Tractor drawn sugarcane cutter planter with discs: It is a tractor mounted implement suitable for cutting and planting sugarcane setts and application of granular fertilizer in single operation The machine accepts whole cane for planting and has been developed by IISR, Lucknow.

Tractor mounted ridger type sugarcane cutter planter: It is a tractor mounted implement suitable for cutting and planting sugarcane setts and application of granular fertilizer in single operation. The machine accepts whole cane for planting. This implement is developed at IISR, Lucknow.

Ring pit digger: In order to facilitate the digging operation, a tractor drawn pit digger is developed at IISR, Lucknow in 1987. The equipment can dig two pits at a time.

Ratoon management device (RMD): It is the device for ratoon management and developed by IISR, Lucknow. It works as stable saving, sub-soiler and ridge making etc.

Animal drawn sugarcane earthing hoe: It is an animal drawn implement suitable for earthing operation of sugarcane crop having row spacing of 90 cm to 120 cm. This implement is developed at MPKV, Pune during 1987-95.

Sprayers and dusters adopted for sugarcane: 1. Wide swath spray boom for tall Crop, 2. Self propelled boom sprayer, 3. Bucket type sprayer, 4. Knapsack Sprayer

Dusters: 1. Bellow typed dusters, 2. Shoulder carried hand operated rotary duster, 3. Sigma dusters, 4. Rotary type hand and power duster

Aerated Steam Therapy: Sugarcane crop often becomes susceptible to sett born diseases. The following methods are used to protect the sett from sett born diseases.

- **i) Hot Water Treatment:** Cane is treated with hot water at 50 o C for 2 hours.
- **ii) Hot Air Treatment:** Cane is treated with hot air at 54 ^{o}C with high humidity for 4 hours.

Tractor mounted 2 row sugarcane stubble shaver: This implement is developed by IISR, Lucknow. The stubble shaver consists of a rotary blade operated by tractor PTO. Use of this equipment results in 15 per cent increase in yield compared to conventional method.

Approximate manufacturer cost (at IISR, Lucknow) of agriculture equipments used in sugarcane cultivation

S. No.	Name of the equipment	Appox cost (Rs.)
1	Culti Harrow	40000
2	Cane Manager	28000
	A. Furrower with Frame	19600
	B. Tynes with Frame	15740
3	Trencher	13800
4	Paired Row Sugarcane Planter	54700

5	**RBS Cane planter**	**115000**
6	**Ratoon Management Device**	**117000**
7	**Detrasher**	**55000**
8	**Harvester**	**100000**
9	**Ratoon Manager**	**82300**
10	**Sugarcane Planters**	**55000-92300**
11	**Three Row Cane Planter**	**61500**

Source: IISR, Lucknow

7.13 Sustainable Sugarcane Initiative (SSI)

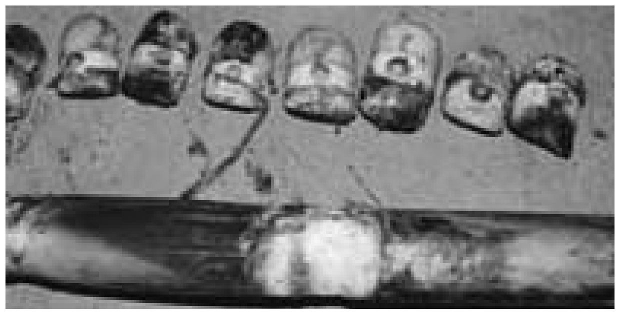
The ICRISAT-WWF project has designed the Sustainable Sugarcane Initiative, (SSI) which is a combination of cane planting innovations and water saving practices that have great potential for not only meeting the growing demands of sugar sector players looking for increased revenues and profitability, but also for the bigger picture of improved natural resource management, reduced environmental footprints and improved livelihoods by means of technologies that are appropriate and effective at household farm level. Sustainable Sugarcane Initiative is a method of sugarcane production that involves the use of less seeds, less water and optimum utilization of fertilizers and land to achieve more yields. Driven by farmers, SSI is an alternative to conventional seed, water and space intensive sugarcane cultivation.

The major principles that govern SSI are:

- Raising nursery using single budded chips.
- Transplanting young seedlings (25-35 days old).
- Maintaining wide spacing (5X2 feet) in the main field.
- Providing sufficient moisture and avoiding inundation of water.
- Encouraging organic method of nutrient and plant protection measures.
- Practicing intercropping for effective utilization of land.

Raising nursery using single budded chips

In the conventional method, 2-3 budded sugarcane setts are used for planting. In SSI, single budded chips, carefully removed from healthy canes are used for raising nursery. The selected buds are placed in trays filled with coco-pith (coconut coir waste) to raise the seedlings. By raising nursery, high percentage of germination can be achieved within a week depending on the agro climatic conditions.



The procedure given below is to be followed for the selection of healthy buds:

- Select healthy canes of 7 to 9 months old which have good internode length (7 to 8 inches) and girth.
- Observe and avoid canes with disease infestation like fungus growth, spots etc.
- Cut the required quantity of canes (refer table 2). Farmers who are unable to go for immediate chipping of buds may keep the cut canes for about a week under shade.
- Remove buds from the selected canes using an implement called Bud Chipper (as shown in the picture). The Bud Chipper comprises a handle and a cutting blade fixed on a wooden plank.
- Keep the cane on the plank and adjust it in such a way that a single bud is placed exactly below the cutting blade. When the handle is pressed, single bud chip comes off the cane.
- Large number of buds (about 150/hr) can easily be chipped off in this way in a short period of time.
- The chipped buds are to be treated with organic or chemical solutions.

Transplanting young seedlings

The young seedlings raised in the nursery are transplanted to the main field at the age of 25 – 35 days. It is important to note here that this one month growth of seedlings achieved under SSI method cannot be achieved even after two months in conventional method.

Transplanting at wider spacing

In conventional methods, the distance between two rows is 45 to 75 cm (1.5-2.5 ft), and 16,000 three budded setts (48,000 buds) are directly planted in the soil to achieve normal population of 44,000 canes per acre. But unfortunately, only 25,000 millable canes are achieved at the end. On the other hand, in the SSI method of sugarcane cultivation, wide spacing of 5X2 feet maintained in the main field leads to 45,000 to 55,000 millable canes because of more tillering. So, wider spacing in SSI cultivation not only reduces the seed usage from 16,000 three budded setts to 4,000 to 5,000 single buds, but most importantly it also supports easy air and sunlight penetration in the crop canopy for better and healthy cane growth.

Water management

In SSI water management, is emphasized so that sufficient moisture is provided rather than inundating the field. Measures like raising of nursery, adopting furrow/alternate furrow irrigation, application of water through drip irrigation etc should be followed. It saves water by 40%.

Organic method of cultivation

The SSI method discourages high application of chemical fertilizers, use of pesticides and weedicides. Farmers should use more organic manures, bio-fertilizers and follow biocontrol measures. The sudden switch over to organic cultivation is not advisable. Instead, a gradual reduction of inorganic and adoption of organic methods can be tried by framers for long term benefits.

Pit System of Planting

In this method, the seedlings are planted in circular pits dug out with specific diameters and distances. The circular pits of 3 ft or 5 ft diameter are dug out to a depth of 1.5 to 2 ft. Row to row spacing is maintained at 7 ft and pit to pit spacing is maintained at 6 ft. At these spacing's, about 1050 to 1150 (with 3 ft diameter) or 500 to 550 (with 5 ft diameter) pits can be made per acre. The pits are then filled with loose dug out soil, FYM or press mud leaving about 1 ft space at the top. Two to four seedlings per pit with 3 ft diameter and 6 to 8 seedlings per pit with 5 ft diameter can be planted close to the edge and covered with soil to a thickness of 5 cm. About 2000 to 4000 seedlings are sufficient per acre, saving the seed cost further for a farmer. All other crop management practices can be followed as practiced in normal method.

Following are the benefits of pit methods:

- In the subtropics and in the tropical part of India, about 25-50 % higher yields were obtained.
- Growth of the crop will be vigorous and the maturity will also be earlier compared to the normal method.

- In case of drip irrigation, nutrition supplied through drip fertigation which will helps in faster crop growth.
- It allows a farmer to pay better attention to the crops or crop pits.
- It gives better ratoon crops and is useful under saline soils and saline water irrigated conditions.
- All the shoots will be of the same age, so there is uniform growth and sugar yield.
- The seedlings are placed at a depth and therefore would moist, hence, in case of drought, or non-availability of water, the yields will not get affected.

Ratoon Management in Sugarcane

It is the most commonly followed and important practice in sugarcane cultivation. In ratoon crops, there is a saving in cost of cultivation in terms of land preparation, seed canes, etc. If ratoons are well maintained, they give high yields. But, for a better ratoon crop, a better plant crop is necessary. Within a week after harvesting the plant crop, ratoon management practices like stubble shaving, off baring, gap filling etc., should be initiated.

Stubble shaving

- The stubbles should be cut using a very sharp blade just above ground level.
- It helps the healthy underground buds to sprout and establish a deeper root system.
- The deeper root system facilitates optimum utilization of the nutrients and moisture available in the lower soil layers and provides good support for growth of the ratoon crop.

Off baring

- It is an operation wherein the ridges are broken or cut on either side using a plough.
- This will loosen the soil to develop better root system and thereby better absorption of nutrients and water.

Gap filling

- If there are no cane clumps for a distance of more than 60 cms or so, it can be considered as a gap. Gap filling can be done using the seedlings raised in the nursery.
- Clumps with excess sprouting can be uprooted, cut into quarters and planted in the gaps.

Row thinning

In areas where close spaced plantings are followed, entire canes of alternate rows can be removed. This can be done by running a plough along the sides of the alternate ridges selected for removal of the cane rows. This will break or loosen the ridges and facilitate easy lifting and removal of the plants. While removing the canes, gap filling in the adjacent rows can be done. This practice of removing alternate rows of canes will increase the space between the rows and thus facilitate sprouting of more tillers because of optimum utilization of the available nutrients and sunlight.

Fertilizer application

- Entire dose of phosphorous, one-third each of nitrogen and potassium as recommended for plant crops can also be applied to ratoon crops. The suggested dose should be applied soon after stubble shaving and off barring, and covered with soil.
- The remaining dose of nitrogen and potassium can be top dressed in equal splits around 30^{th} and 60th days.

Besides the above mentioned practices, all the other crop management practices like irrigation, weeding and earthing up should be continued and followed as done for plant crops. Ratoon crops mature one month prior to the plant crops. In the conventional method of sugarcane cultivation, ratoon crops are maintained for only a maximum of two seasons but farmers practicing SSI methods maybe able to achieve 5 to 6 ratoon crops.

Chapter 8: Cropping system

Sugarcane prevailing cropping system in India:

Cropping system for Sub tropical	Cropping system for Tropical region
Paddy- Autumn Sugarcane-ratoon-wheat	Bajra-Sugarcane(pre-seasonal)-Ratoon-wheat
Greengram- Autumn Sugarcane-ratoon-wheat	Paddy-Sugarcane-Ratoon- Finger millet
Maiz- Autumn Sugarcane-ratoon-wheat	Paddy-Sugarcane-Ratoon- Wheat
Kharif Crops-Potato-Spring Sugarcane-ratoon-Wheat	Paddy-Sugarcane-Ratoon- gingelly
Kharif Crops-Mustard-Spring Sugarcane-ratoon-Wheat	Paddy-Sugarcane-Ratoon- urd.
Kharif Crops-Pea/Coriander-Spring Sugarcane-ratoon-Wheat	Cotton-Sugarcane-Ratoon–wheat
Kharif Crops-Wheat-late Planted Sugarcane-ratoon-Wheat	Sugarcane-Ratoon-Kharif rice-Winter rice.

The problem poses by the prevalent cropping systems is mainly depletion of ground water, due to continuous adoption of cropping system having no time for summer ploughing resulted in incidence of white grub in sugarcane emerging as serious problem.

Chapter 9: Crop products

9.1 Products and by- products of Sugarcane

Sugarcane based Sugar industry is one of the largest and important industry in tropical and sub tropical countries of the world. The Sugarcane plant offers a huge potential, not only as the sucrose of a very important food but also as a source of energy and valuable commercial products from fermentation and chemical synthesis. Sugarcane processing is focused on the production of cane sugar from sugarcane. Sugarcane is considered as one of the best converters of solar energy into biomass and Sugar. Sugarcane is a rich source of food (Sucrose, jiggery andsyrups), fiber (cellulose), fodder (green top, bagasse, molasses) fuel and chemicals (Bagasse molasses & alcohol). During the process of sugar production, the main by product of cane sugar industry are Bagasse, Molasses and Press mud. The other co products and by products of less commercial value are Green leaves, green tops, trash, Boiler ash and effluents generated by sugar industry and distillery. There are many other industries which are based on sugarcane by diversification and utilization of co-products and by products of the sugar industry, instead of merely depending on production of sugar. Thus the effort should be for integral utilization of sugarcane, its coproducts and by products to produce many value added products, to derive maximum benefits from sugarcane crop.

9.2 Bagasse based industries:

Bagasse is a fibrous residue left over after the sugarcane is crushed in the milling plant of the sugar factories for extracting its juice. Bagasse consists of water, fiber and relative small quantity of soluble solids. The fibre content of bagasse (cellulose) is used in cellulosic industries like pulp plant, paper plant, particle boards industries, co generation unit using bagasse as a fuel, cattle feed from bagasse, cultivation of edible mushroom on bagasse, production of furfural from bagasse etc.

9.3 Molasses based industries:

Molasses is another important by product of the sugar industry. It is the mother liquor left over after the crystallization of sucrose from which further quantity of sucrose can not be recovered economically. The yield of molasses per tone of cane may vary within a range of 2.2- 3.7%. Molasses contains about 30-35% of sugar and 15-20% of reducing sugar (Glucose and fructose) and thus the total sugar content of molasses is 45-55%. It is by virtue of its total sugar content the molasses is a valuable raw material for the production of many value added products. The main products that cane be produced from molasses are (i) Distillery (ii) Acetic Acid (iii) Fuel Alcohal (iv) Bio gas from effluent treatment (v) Cattle feed (vi) Ethyl Alcohal (vii) Bakers yeast (viii) Lactic Acid (ix) Citric Acid (x) Glycerol (xi) Butanol- Acetone (xii) Monosodium Glutamate (xiii) Ephedrrinee Hydrochloride as Pharmaceutical use etc.

9.4 Press mud based industries:

Press mud is a rich source of organic carban and contains a good proportion of N, P, Ca, Fe & Mn. In early stage the disposal of press mud is posing a problem before the sugar factories not only related to the volume to be handled but also to its polluting effect and an increase in population of insects such as house flies etc. Now, due to technological advancement the press mud is largely utilized as a fertilizer and in the wax and compost industries. Uses of press mud are fertilizer, animal feed, Cane wax, Bio-gas, steam generation cake and mud cake in building materials.

Chapter 10: Crop Development Programme

10.1 Brief description on each important crop development proramme

Since October 2000, the Crop oriented Centrally Sponsored Scheme being implemented in different States subsumed with Macro Management Mode on Agriculture (**MMMA)**. **The pattern of assistance has been 90:10 between GOI and State Govt.** The main components continued to remain same as it was in the SUBACS **with the flexibility to incorporate new components as per the requirement of the state.** The main components of the scheme are: Field demonstration, IPM demonstration, Farmers Training, State level training, Distribution of implements (Manual/tractor drawn), Seed multiplication, MHAT Plant, Drip irrigation infrastructure, Bio-fertilizer distribution, Biological control etc.

Components and pattern of Assistance in Sugarcane Development Programme:

S.No	Components/ interventions	Unit	Pattern of Assistance
1	Demonstration of Technology	No.	0.5 ha. @ Rs. 7500/- per demo.
2	Distribution of Farm Implementation	No.	i) Bullock/Manual Drawn @ 25% of the cost limited to Rs. 2500/- per unit. ii) Tractor/Power drawn @ 25% of the cost limited to Rs.15000/-
3	Multiplication of Planting Materials(Cane sets)	Ha.	a) Foundation nursery @ 10% of the cost limited to Rs.4000/- per ha. b) Primary nursery @ 10% of the cost limited to Rs.2,000/- per ha.
4	Training	No.	a) Farmers: 50 farmers for 2 days @ Rs.10000/- per training. b) State Level: 30 participants for 3 days @ Rs. 20,000/- per training.
5	Setting up of Moist heat treatment units for treatment of planting material by Farmer's Association, Mills	No.	@ 50% of the cost per plant including generator, limited to Rs. 3,00,000/-.
6	Supply of Drip irrigation Infrastructure	Ha.	@ 50% of the cost limited to Rs. 30,000/-. per ha.
7	Setting up/Strengthing up a) Tissue Culture Lab b) Bio Agent Lab by SAU /ICAR Instt., Mills	No.	@ 25% of the cost limited to Rs. 10,00,000/-. per tissue culture lab or bio agent lab.
8	Monitoring/Inspection/visit/ Preparation of reports, POL etc. as contingency	Distt.	Rs. 50,000/- lump sum per major sugarcane growing district.

9	Area specific intervention		10% of allocation
	NEW COMPONENTS		
10	Assistance for Boring of Tubewells/pumpsets	No.	@25% of cost limited to Rs. 12,000/- per set.
11	Assistance for distribution of Micro Nutrients	Ha.	@25% of cost limited to Rs. 1000/- per ha.
12	Distribution of Planting Matterial and Soil Treatment Chemicals	Ha.	@25% of cost limited to Rs. 1000/- per ha..
13	Visit of Farmers to model farms, institutes etc.	Nos.	40 sugarcane farmers @ 50% of the cost limited to Rs. 50,000/-

In Sugarcane Development Programme under Macro Management Mode during 2011-12 the total financial target was Rs. 2871.88 Lakh and achievement was Rs. 2779.79 Lakh by different state. Component wise progress given at ***Annexure – XXXII***.

Chapter 11: Special initiatives taken for encouraging the cultivation of the sugarcane crop

11.1 Minimum support price (MSP) including additional incentives (Bonus) on procurement of crop produce from the states:

The Minimum Support Price (MSP) or Fair and Remunerative Price (FRP) in sugarcane is declared by Govt. of India every year to protect the benefits of the sugarcane growers keeping in view the highest of the inputs and other parameters. The State Government also declared State Agreed Price for Sugarcane over the FRP. The minimum statutory price/ fair & remunerative price of sugarcane fixed by the government is given at ***Annexure-XXXIII***

Chapter 12: Workshops, conference and seminars

12.1 Outcomes/ recommendations of annual workshop conducted by ICAR/ SAUs:

Recommendations pertaining to Sugarcane crop emerged during National Conference for *Kharif* Campaign-2009 held on 20th & 21st March 2009.

S.No	Recommendations made during Kharif campaign 2009
1.	Adoption of IPM measures against wooly aphid in Maharashtra and Karnataka
2.	Encourage cultivation of proper intercrops to increase returns to the farmers.
3.	Adopt efficient irrigation techniques to achieve higher water use efficiency; through use of drip irrigation.
4.	Effort to increase area under early maturing varieties with high recovery percentage.

12.2 Recommendation of important conference / seminar / brain storming sessions related to crop.

a. Recommendations emerged during Pre-*Kharif* DAC-ICAR Interface held on 16th February 2006.

Issue	Recommendation made by ICAR
Development of sugarcane varieties resistant to flooding particularly for Eastern India.	Water logging tolerant varieties such as Jalpari (CoSe 96436), Sarayu (Co 87263), Moti (Co 87268), Gandak (Co 89029), Pramod (BO 128) have been developed for water logging areas of Eastern India, Breeder Seed production is taken up as per the indent received from DAC.

b. Recommendations emerged during Pre-*Rabi* DAC-ICAR Interface held on 30-31 August 2005

<table>
<tr><th>Issue</th><th>Recommendation made by ICAR</th></tr>
<tr><td>In spite of very low productivity of sugarcane, Chhattisgarh State is encouraging the cultivation of crop and has also established sugar mills. Suitable sugarcane varieties for Chhattisgarh with their yield potential may be indicated.</td><td>Chhattisgarh is not a traditional sugarcane growing State. However, following varieties developed for peninsular zone may be evaluated in Chhattisgarh State.
<table>
<tr><th>Variety</th><th>Cane Yield</th></tr>
<tr><td>Early</td><td></td></tr>
<tr><td>Co 85004 (Prabha)</td><td>90.5</td></tr>
<tr><td>Co 94008 (Shyama)</td><td>119.8</td></tr>
<tr><td>Midlate</td><td></td></tr>
</table></td></tr>
</table>

		Co 86032 (Nayan)	102.0	
		Co 87025 (Kalyani)	98.2	
		Co 87044 (Uttara)	101.0	
		Co 8371 (Bhima)	117.7	
		CoM 88121 (Krishna)	88.7	
		Co 91010 (Dhanush)	116.0	

c. Recommendations emerged during Pre-kharif DAC-ICAR Interface held on 19th February 2009

Issue	Recommendation made by ICAR
1. Development of IPM Module for the control of White Grub in sugarcane (emerging as a serious problem in western UP in last two year	1. IPM Module for the control of White Grub in sugarcane involves; **Mechanical Measure:** i. Mass collection of adults beetles through light trap after first shower in May-June at 7-8 PM ii. Deep ploughing before first shower in May **Cultural Measure:** - Crop rotation with paddy - Flooding of infested field for 5 days (5 cm water level) during July-August Pheromone Dispener: -Select one host tree within a radius of 15 m and spray monocrotophos 36 WSC or quinalphos 25 EC, 0.05% during day time and in evening, place 3-4 pheromone dispensers/three continuously for 3-4 evenings after adult emergence -Drenching of chlorpyriphos 20 EC , quinalphos 25 EC @ 4 L /ha or Imidacloprid 200 SL @ 300 Ml /ha in 1500 L water in standing crop within 21 days of mass emergence of beetles or soil application of phorate 10 G or quinalphos 5 G @ 2 kg ai per ha at plant base during 1st week of July.
2. Development of sugarcane harvester suited for topographical area to reduce the cost of production and timely harvesting of sugarcasne	Tractor operated front mounted two raw sugarcane harvester has been development at IISR, Lucknow for basal cutting. A separate equipment detrasher cum detopper has also been developed for de-trashing of the harvested sugarcane stalks. Both these equipment can be used in conjunction as a system for sugarcane harvesting, The system is suitable for prevalent agronomic practices in vague in the country.

d. Recommendations emerged during Pre-kharif DAC-ICAR Interface held on 11th February 2010

<table>
<tr><th>Issue</th><th>Recommendation made by ICAR</th></tr>
<tr><td>To develop a technology package for enhanging sprouting in winter harvested cane.</td><td>For overcoming this important constraints to productivity of ratoon crops, pre harvest application of the potassium (80 kg K2O/ha) with last irrigation one month before harvesting and foliar spray of 200 ppm ethrel 15 days before harvest, improved sprouting and productivity of winter initiate ratoon. Spraying of Glyphosate @ 0.16 kg a.i per ha on to the standing crop 2 months before harvesting improve not only the juice quality of the crop but alos improves sprouting of the stubble buds in the succeeding ratoon crop. Post harvest application of Cycocel @ 5 kg ai per ha increased the yield of succeeding ratoon crop by 23%. Post harvest application of the formulation containing Vitamins and Minerals, and potassium also improves sprouting of suitable buds as well as productivity by 22-23% in winter initiated ratoon.</td></tr>
<tr><td>2. Development of sugarcane variety suited for water logging condition.</td><td>A number of sugarcane varieties have been released and notified for cultivation for different sugarcane growing zones of the country including states of UP and Bihar. List of sugarcane varieties tolerant to water logging are given below”
<table>
<tr><th>S. No.</th><th>Sugarcane growing zones</th><th>Variety</th><th>Remarks</th></tr>
<tr><td rowspan="2">A.</td><td rowspan="2">North west zone (Punjab, Haryana, Rajasthan, Central and western UP and Uttrakhand)</td><td>Co-98014 (Karan-1)</td><td>Early</td></tr>
<tr><td>Co-pant 97222</td><td>Midlate</td></tr>
<tr><td rowspan="4">B.</td><td rowspan="4">North central zone (Eastern UP, West Bengal and Jharkhand)</td><td>Cose-96436 (Jalpari)</td><td>Midlate</td></tr>
<tr><td>Colk-94184 (Birendra)</td><td>Early</td></tr>
<tr><td>Co-0233 (Kamal)</td><td>Early</td></tr>
<tr><td>Co-0233</td><td>Midlate</td></tr>
</table>
</td></tr>
</table>

12.3 Brain Storming on 24th March, 2009 at IISR, Lucknow:

A brain storming discussion was held in IISR, Lucknow to improve sugarcane production and sugar recovery in India with emphasis on UP and Maharashtra under the Chairmanship of Dr. S.P. Tiwari, DDG (CS & Edn.) on 24th March 2009. In the session the following points were set for low productivity and sugar recovery:

i. Monoculture of sugarcane i.e. lack of crop rotation in some areas, leads to deletion of nutrients in soil and adversely affect cane productivity.
ii. Cultivation of rejected and unapproved varieties not only lealds to low cane productivity, but also pose a risk to buildup of diseases, particularly red rot.
iii. Ratoon crop is generally receives much less attention and care by the cane growers leading to lower cane productivity.
iv. Post harvest deterioration in cane quality on account of staling and delayed crushing contribute to low sugar recovery.
v. Inadequate availability of quality seed of new sugarcane varieties and poor seed replacement rate adversely affect the realization of potential cane yield of varieties.
vi. Increasing problem of soil compaction/ hard pan coupled with inadequate sub-soiling not only adversely affects plant growth, but also promotes lodging of canes.
vii. In U.P. rainfall in the month of May and June 2008, coinciding with the tillering phase in sugarcane, adversely affect tillering which led to lesser number of millable canes. Due to continuous rains, fertilizers could not be applied on time. These factors coupled with shortage of phosphatic and potassic fertilizers, particularly in UP, led to low cane productivity.
viii. Relatively higher minimum temperature and higher relative humidity during maturity phase delayed sugarcane ripening resulting in low sugar recovery. Moreover, crushing of immature cane due to imbalance crushing schedule lowered sugar recovery.
ix. In Maharashtra, low rainfall and inadequate water availability during grand growth phase together with shortage of fertilizer (DAP), labour and power resulted in reduced cane productivity.
x. Area under early maturing high sugar varieties is very less which led to low average sugar recovery.

12.4 Recommendations

To overcome these reasons following recommendation have been made:

a. Researchable issues:

i. While characterizing new sugarcane varieties, POL (%) cane may also be reported to better indicate its recoverable sugar percent.
ii. Weather based fore-warning system may be developed for management of major insect pest and diseases of sugarcane.
iii. Work on sugarcane ripeners may be intensified at location where conditions are not favourable for ripening. The Bio-safety aspects of the chemicals used as ripeners should also be duly addressed.

b. Developmental issues:

i. Seed production of recommended varieties of sugarcane should be intensified by the State Govt. department and three tier seed programme be strictly followed. Micro-propagation (tissue culture) may be made an integral part of seed production programme for quick multiplication of disease-free seed material. In seed production programme, molecular disease indexing techniques developed at research institutes may be employed for monitoring of healthy seed production.

ii. IISR, Lucknow may circulate a list of varieties available for seed production to sugar factories as well as to ISMA and NFCSF to enable sugar factories to send indent of seed to IISR.

12.5 Recommendations of the National Seminar on "Mechanization of Sugarcane Cultivation" held at IISR, Lucknow on March 19-20, 2010.

After deliberations, following recommendations/ Action Plan emerged for spread of mechanization in sugarcane cultivation:

i. Plan to be prepared by the sugar mills for mechanical planting in their respective area in the initial stage and subsequently more operation could be included.

ii. Identifying the machine and source of supply in consultation with IISR, Lucknow, if required.

iii. Manufacturer selected should ensure the quality control for which he shall send prototype to IISR for performance testing, if desired. This quality will have to be maintained by him in future supply.

iv. Manufacturer will provide working design of the supplied equipment and will ensure after sale service during the use period.

v. IISR may provide training to the manufacturers and the users at their cost for the proper operation and maintenance of the equipment.

vi. Any suitable device pertaining to the sub-soiling through RMD/Sub-soiler should be used to improve soil health and ratoon productivity. For this also sugar factories should plan in advance.

vii. Initially, DSCL and Balrampur group of Industries have agreed to initiate this plan in their areas of operations.

viii. Group meeting may be arranged in future to take stock of the progress made in this regard.

12.6 Important Institution

a. National

i. Vasantdada Sugar Institute Manjari (Bk), Tal. Haveili, Dist. Pune, PIN-412 307 (Maharashtra)

ii. Indian Institute of Sugarcane Research, Rae Bareli Road, Post Dilkusha, Lucknow-226 002 (U.P.)

iii. Sugarcane Breeding Institute, ICAR Coimbatore-641 007 (Tamilnadu).

iv. National Sugar Institute, PO. NSI, Kalyanpur, Kanpur – 208017 (U.P.)

v. U.P. Council of Sugarcane Research Sahjahanpur, U.P.

vi. Indian Sugar Mills Association, Ansal Plaza, 'C' Block, 2nd Floor, August Kranti Marg, Andrews Ganj, New Delhi- 110049

vii. National Federation of Cooperative Sugar Factories Ltd (NFCSF) Ansal Plaza, Block-C, 2nd Floor, August Kranti Marg, New Delhi.

b. International:

i. International Society of Sugar Cane Technologists, Mauritius,

ii. Brazilian Society of Sugar and Ethanol Technologists, Brazil.

c. National and international Agencies informative for farmers, development officers, policy planners and scientific community.

Indian Sugar Mills Association (ISMA)
Sugar House, 39, Nehru Place,
New Delhi-110 019.
E-mail: sugarmill@nda.vsnl.net.in

National Federation of Co-operative Sugar Factories Ltd., (NFCSF)
Vaikunth (IIIrd Floor), 82-83, Nehru Place, New Delhi-110 019.
E-mail: nfcsf@ndb.vsnl.net.in

All India Distillers Association (AIDA)
805, Siddharth, 96, Nehru Place, New Delhi 110 019.
E-mail: distiler@vsnl.net.in

The Sugar Technologies Association of India (STAI)
21, Community Centre, East of Kailash, New Delhi 110 065.
E-mail: staidel@vsnl.net.in

Indian Sugar and General Industry Export Import Corporation Ltd.,
21, Community Center, East of Kailash, New Delhi 110 065.
E-mail: isiec@ndb.vsnl.net.in

Sugar Technology Mission
D-5 Apartment, Qutab Hotel, New Mehrauli Road, New Delhi 110 016.

d. Website of advisory services to farmers: The website developed by the Sugarcane Department, Uttar Pradesh www.up.cane.org. This website is providing sugarcane advisory, their marketing, supply ticket, weighment, cane payment to the farmers. This website is very helpful for the sugarcane growers I the state of Uttar Pradesh.

http://www.upcane.org/sis/en/index.asp

http://www.sugarindia.com/glossary.htm

http://www.vsisugar.com

http://www.fcamin.nic.in/dfpd_html/index.asp

http://www.cogenindia.org/default.asp

http://www.staionline.org

http://ncdc.nic.in/index.php

http://www.coopsugar.org

http://www.indiansugar.com

http://www.tifac.org.in/index1.htm

http://nsi.gov.in

www.sugarcaneweb.co.uk

www.bonsucro.com

www.issct2013.com.br

www.sugarcane.res.in

www.wsro.org

www.upcane.org

www.sugarcaneweb.co.uk

www.vsisugar.com

www.sugarbazar.com

http://www.issct.org

12.7 Researchable and Developmental issues:

For crop specific research issues like e.g. varieties, production technologies farm implements, climate change and product development, value addition, marketing etc.

a. **Researchable issue:**

i. Development of the varieties capable of giving higher cane yield, sugar recovery along with field stability and good Ratooning ability
ii. Assessment and refinement of agro-techniques for sustainable farming system and management of sugarcane under late planting situation.
iii. Development of module for controlling whit grub and mealy bug in sugarcane which is emerging as a serious problem.
iv. Designing, developing, Sugarcane harvester suited in Indian topographical, cultural and economical conditions.
v. Enhancement of sprouting in winter harvested cane.

b. Developmental issue:

i. Strengthening of seed production programme at institute or research centre Simultaneously focusing of rearing of quality seed nurseries at farmers field.

ii. Cluster approach in Transfer of technology with modern tools.

12.8 Strategies for sugarcane productivity enhancement:

a. Agronomic Strategies:

- Adoption of suitable varieties & their blending.
- Strengthening of seed production programme
- Water Management.
- Integrated Nutrient Management Approach.
- Integrated Pest Management.
- Ratoon Management.
- Adoption of suitable Time and Method of Planting.
- Promotion of intercropping.
- Promotion of mechanization.
- Credit flow and its utilization.

b. Extension Strategies:

Timely and efficient dissemination of the innovative crop production technologies to the farmer's field is very essential for increasing sugarcane productivity through modern tool of TOT:

- Demonstration.
- Training programmes at different level.
- Exposure visits.
- Distribution of technical literature.
- Village *Gosthies*
- Farmers Field School (FFS)
- Audio-visual aids/video clippings.

Since, the development is a regular feature and joint venture and It should be made with the help of concerned department of State Government extension officials, sugar factories, SAUs, KVK's, Regional *Krishak Sewa Kendra* etc.

c. Diversification & Automation in sugar industry:

Sugarcane is a multi-product crop and therefore, avenue for use of its by-products and co-products is advocated to make the sugar industry economically viable consistently and making the sugarcane cultivation sustainable through assured marketing. The following diversification in the sugar industry is advocated by establishing the bi-product based subsidiaries:

i. **Baggase based:** Co-generation, Paper, News print, MDPB etc

ii. **Molasses based:** Ethanol, Distilleries, cattle feed , pharmaceutical,

iii. **Press mud bases:** Bio-compost

iv. **Other value added product**s etc

d. Development need:

i. To popularize techniques for conservation of water which includes trash mulching, cultivation of sugarcane varieties with deep root system, successful application of water under skip furrow method.

ii. Inter-cropping, green manuring and dual purpose legumes need to be strengthened and popularize to meet the triple fold objectives of nutrient economy, crop diversification and mid season income generation.

iii. There exist an ample scope of reducing the production cost of sugarcane, enhance profitability and regulate the cane supply to sugar mills for maintaining sugar recovery in sugar mills through ratoon canes, therefore, multiple Ratooning with proper ratoon management practices is advocated.

iv. Recycling of sugarcane crop residues viz. root/trash through vermi-culture, press mud as a soil ameliorants from sugar factory and distilleries is promoted for sustainability of soil.

v. Balance varietal planning with adequate proportion of early, mid-late varieties in each sugar factory area is required, so that, sugarcane with the maximum sucrose content is supplied throughout the crushing period to enhance the realize sugar recovery.

vi. Possibility may be explored for pre-harvest maturity survey, harvesting schedule and supply management in a proper sequences to sustain the sugar recovery.

vii. Selection of cane seed and its preparation before planting is highly desirable to avoid the incidence of different diseases like Red rot, wilt, smut, etc.

e. Recommendations of Dr. C. Rangarajan committee Report

i. Prime Minister had set up a committee under the chairmanship of **Dr. C. Rangarajan, Chairman, Economic Advisory Council** to the Prime Minister to look into all the issues relating to the deregulation of the sugar sector. The committee has completed its task, after several rounds of deliberations, consultations with stakeholders, and discussion with Chief Ministers of major sugar-producing states. The report was submitted to the Prime Minster on 10-10-2012.

ii. A major recommendation of the committee relates to revising the existing arrangement for the price to be paid to sugarcane farmers, which suffers from problems of accumulation of arrears of cane dues in years of high price and low price for farmers in other years. The existing arrangement comprises a Fair and Remunerative Price (FRP) announced each year by the Centre, under the Sugarcane Control Order and on the advice of CACP, as the minimum price of sugarcane. However, many states in north India also announce a State Advised Price (SAP) under state legislation. Generally, the SAP is substantially higher than the FRP, and wherever SAP is declared, it is the ruling price. Instead of the present arrangement, the committee has proposed that at the time of cane supply, farmers be paid FRP as the minimum price, as at present. Further, subsequently, on a half-yearly basis, the state government concerned would announce the *ex*-mill prices of sugar and its by-products, and farmers would be entitled to a 70% share in the value of the sugar and by-products produced from the quantity of cane supplied by each farmer. Based on the share so computed, additional payment, net of FRP already paid, would then be made to the farmer. Since the sugar value estimate includes return on capital employed, this implies that farmers would also get a share of the profits. With such a system in operation, states should not declare an SAP.

iii. The committee has also recommended dismantling of the levy obligation for sourcing PDS sugar at a price below the market price. States should be allowed henceforth to fix the issue price of PDS sugar, while the existing subsidy to states for PDS sugar transport and the difference between the levy price and the issue price would continue at the existing level, augmented by the current level of implicit subsidy on account of the difference between the levy price and the open market price. This will free the industry from the burden of a government welfare programme, and indirectly benefit both the farmer and the general consumer since the industry passes on the cost of levy mechanism to farmers and consumers.

iv. The committee has recommended dispensing with the present mechanism of regulated release of non-levy sugar, as it imposes additional costs on factories on account of inventory accumulation.

v. The committee has recommended that cane area reservation ultimately be phased out and contracting between farmers and mills allowed for enabling the emergence of a competitive market for assured supply of cane, in the interest of farmers and economic efficiency. However, in case some states want to continue it for the time being, they should do so while ensuring that area reservation is done for at least three to five years at a time, so that industry has a stake in its development. Further, wherever and whenever a state discontinues area reservation, the Centre should remove the stipulation of a minimum distance between two mills.

vi. On external trade, the committee has favoured a stable policy regime with modest tariff levels of 5% to 10% ordinarily, and dispensing with outright bans and quantitative restrictions. The committee has also recommended dispensing with the mandatory requirement of jute packaging. In respect of molasses, the committee favours free movement and dismantling of end-use based allocation quotas that are in vogue in several states, to enable creation of a national market and better prices for this valuable by-product as well as improved efficiency in its use.

vii. Members on the committee were Shri T. Nandakumar, Member, NDMA, Dr Ashok Gulati, Chairman, Commission on Agricultural Costs & Prices, Dr Raghuram Rajan, Chief Economic Advisor, Shri Sudhir Kumar, Secretary, Food & Public Distribution, Shri Ashish Bahuguna, Secretary, Department of Agriculture & Cooperation, and Dr K. P. Krishnan, Convener, as the then Secretary, Economic Advisory Council.

Annexure-I(a)

State wise Area of sugarcane crop in India

(Lakh ha)

Sl.No.	State	2008-09	2009-10	2010-11	2011-12	2012-13*
	A. Tropical					
1	Andhra Pradesh	2.00	1.58	1.92	2.04	2.00
2	Gujarat	2.20	1.54	1.90	2.02	2.03
3	Karnataka	2.80	3.37	4.23	4.30	4.10
4	Madhya Pradesh	0.71	0.60	0.65	0.69	0.88
5	Maharashtra	7.70	7.56	9.65	10.22	9.40
6	Tamil Nadu	3.10	2.93	3.16	3.82	3.33
	B. Sub-tropical					
7	Bihar	1.10	1.16	2.48	2.35	2.35
8	Haryana	0.90	0.74	0.85	0.95	1.07
9	Punjab	0.81	0.60	0.70	0.80	0.84
10	Uttar Pradesh	20.80	19.77	21.25	21.62	22.77
11	Uttarakhand	1.10	0.96	1.07	1.08	1.12
12	West Bengal	0.20	0.14	0.15	0.16	0.18
	Others	0.78	0.78	0.84	0.80	0.94
	Grand Total	**44.20**	**41.73**	**48.85**	**50.85**	**51.01**

*** Provisional**

Annexure I(b)

State wise production of sugarcane crop in India

(Million tonnes)

S.No.	State	2008-09	2009-10	2010-11	2011-12	2012-13*
	A. Tropical					
1	Andhra Pradesh	15.38	11.71	14.96	16.73	15.90
2	Gujarat	15.51	12.40	13.76	14.17	14.21
3	Karnataka	23.33	30.44	39.66	38.81	32.72
4	Madhya Pradesh	2.98	2.54	2.67	2.68	4.22
5	Maharashtra	60.65	64.16	81.89	81.86	61.32
6	Tamil Nadu	32.80	29.75	34.25	39.28	34.93
	B. Sub-tropical					
7	Bihar	4.96	5.03	12.76	12.07	11.58
8	Haryana	5.13	5.34	6.04	6.95	7.59
9	Punjab	4.67	3.70	4.17	4.67	5.04
10	Uttar Pradesh	109.05	117.14	120.55	128.82	135.64
11	Uttarakhand	5.59	5.84	6.50	6.59	6.85
12	West Bengal	1.64	1.00	1.13	1.17	1.62
	Others	3.34	3.26	4.04	3.87	3.71
	Grand Total	**285.03**	**292.31**	**342.38**	**357.67**	**335.33**

*** Provisional**

Annexure-I(c)

State wise productivity of sugarcane crop in India

(tonnes/ha)

Sl.No.	State	2008-09	2009-10	2010-11	2011-12	2012-13*
	A. Tropical					
1	Andhra Pradesh	82.15	74.11	77.92	82.00	79.50
2	Gujarat	73.03	80.52	72.42	70.20	70.00
3	Karnataka	88.65	90.33	93.76	90.26	79.80
4	Madhya Pradesh	42.40	42.33	41.08	38.70	47.80
5	Maharashtra	81.29	84.87	84.86	80.10	65.20
6	Tamil Nadu	108.15	101.54	108.39	102.80	105.00
	B. Sub-tropical					
7	Bihar	32.44	43.36	51.45	51.50	49.40
8	Haryana	63.29	72.16	71.06	73.30	71.00
9	Punjab	60.27	61.67	59.57	58.40	60.00
10	Uttar Pradesh	57.12	59.25	56.73	59.60	59.60
11	Uttarakhand	62.02	60.83	60.75	61.02	61.20
12	West Bengal	74.71	71.43	75.33	73.13	90.00
	All India	**64.60**	**70.00**	**70.10**	**70.30**	**65.80**

*** Provisional**

Source: Cooperative Sugar, vol-44, No.4, Dec, 2012.

State-wise & District –wise Area, production and yield of sugarcane may be seen at Directorate website: http://dsd.dacnet.nic.in

Annexure-II

State wise sugar production (2002-03 to 2011-12) (000 tonnes)

S. No	State	2002-03	2003-04	2004-05	2005-06	2006-07	2007-08	2008-09	2009-10	2010-11	2011-12*
1	A.P.	1210	886	982	1236	1680	1335	593	515	1006	1135
2	Bihar	408	274	253	422	451	336	214	258	385	450
3	Chhattisgarh	4	17	10	18	24	38	13	9	23	36
4	Goa	13	10	8	11	19	15	9	8	13	10
5	Gujarat	1252	1066	797	1168	1425	1366	1012	1189	1235	1000
6	Haryana	635	582	100	409	652	599	229	248	392	494
7	Karnataka	1868	1116	1040	1943	2662	2900	1651	2558	3683	3872
8	Kerala	2	-	-	-	-	-	-	-	-	-
9	M.P.	71	93	72	94	180	174	60	80	165	159
10	Maharashtra	6219	3175	2217	5197	9100	9075	4578	7067	9054	8977
11	Orissa	40	41	44	40	61	63	31	23	45	65
12	Puducherry	34	20	18	28	60	51	17	19	47	64
13	Punjab	586	390	315	338	486	534	242	181	302	390
14	Rajasthan	2	9	4	6	7	6	4	4	4	2
15	Tamilnadu	1644	921	1108	2142	2539	2141	1597	1280	1846	2379
16	U.P.	5651	4552	5037	5784	8475	7319	4064	5179	5887	6974
17	Uttarakhand	498	387	381	426	535	400	223	292	302	331
18	W.B.	8	7	5	5	8	5	2	2	5	5
	All India	**20145**	**13456**	**12691**	**19267**	**28364**	**26357**	**14539**	**18912**	**24394**	**26343**

*** Provisional**

Source: Cooperative Sugar, vol-44, No.4, Dec, 2012.

Annexure- III

Area, Production and Yield of Sugarcane in India

Year	Area under Sugarcane (000 ha.)	Production of sugarcane (000 tonnes)	Yield of cane per ha. (tones)
1930-31	1176	36354	30.90
1940-41	1617	51978	32.10
1950-51	1707	54823	32.10
1960-61	2415	110001	45.50
1970-71	2615	126368	48.30
1980-81	2667	154248	57.80
1990-91	3686	241045	65.40
2000-01	4316	295956	68.60
2001-02	4411	297208	67.40
2002-03	4520	287383	63.60
2003-04	3938	233862	59.40
2004-05	3662	237088	64.80
2005-06	4201	281172	66.90
2006-07	5151	355520	69.00
2007-08	5055	348188	68.90
2008-09	4415	285029	64.60
2009-10	4175	292302	70.05
2010-11	4885	342382	70.09
2011-12*	5085	357667	70.30

- Provisional
- Source: Cooperative Sugar, vol-44, No.4, Dec, 2012.

Annexure-IV

Area, production and yield of sugarcane in major growing countries (2006 to 2010)

Country	Area (Lakh ha)					Production (Lakh tones)					Yield (Ton/ha)				
	2006	**2007**	**2008**	**2009**	**2010**	**2006**	**2007**	**2008**	**2009**	**2010**	**2006**	**2007**	**2008**	**2009**	**2010**
Argentina	3.15	3.20	3.60	3.45	3.50	264.50	239.60	269.60	255.80	250.00	83.97	74.88	74.89	74.14	71.43
Australia	4.15	4.09	3.81	3.91	4.05	371.28	363.97	326.21	302.84	314.57	89.47	89.07	85.72	77.40	77.67
Brazil	63.56	70.81	81.40	86.18	90.77	4774.11	5497.07	6453.00	6916.06	7174.62	75.12	77.63	79.27	80.26	79.04
China	13.89	15.97	17.54	17.08	16.95	933.06	1137.32	1249.18	1162.51	1114.54	67.18	71.23	71.22	68.08	65.75
Colombia	4.10	4.10	3.83	3.80	3.80	384.50	385.00	385.00	385.00	385.00	93.77	93.86	100.42	101.45	101.32
India	42.01	51.50	50.55	44.20	41.70	2811.72	3555.20	3481.88	2850.29	2923.00	66.93	69.03	68.88	64.49	70.10
Mexico	6.80	6.90	6.69	7.11	7.04	506.76	520.89	511.07	494.93	504.22	74.53	75.44	76.37	69.65	71.63
Pakistan	9.07	10.29	12.41	10.29	9.43	446.66	547.42	639.20	500.45	493.73	49.23	53.20	51.49	48.62	52.36
Philippines	3.92	3.83	3.98	4.04	3.63	315.50	320.00	340.00	325.00	340.00	80.43	83.56	85.43	80.45	93.71
Thailand	9.42	9.86	10.29	9.32	9.78	476.58	643.66	735.02	668.16	688.08	50.57	65.26	71.41	71.66	70.36
Others	47.28	47.58	48.2	48.81	45.99	2939.12	2950.95	2950.81	3009.23	2757.29	62.16	62.02	61.22	61.65	59.95
World	**207.35**	**228.13**	**242.3**	**238.19**	**236.64**	**14223.8**	**16161.08**	**17340.97**	**16870.27**	**16945.05**	**68.60**	**70.84**	**71.57**	**70.83**	**71.61**

Source: FAO.org

Annexure-V

Yield gap of sugarcane in India with other countries

(t/ha)

country	2006	Gap	2007	Gap	2008	Gap	2009	Gap	2010	Gap
Argentina	83.97	17.04	74.88	5.85	74.89	6.01	74.14	9.65	71.43	1.33
Australia	89.47	22.54	89.07	20.04	85.72	16.84	77.40	12.91	77.67	7.57
Brazil	75.12	8.19	77.63	8.60	79.27	10.39	80.26	15.77	79.04	8.94
China	67.18	0.25	71.23	2.20	71.22	2.34	68.08	3.59	65.75	-4.35
Colombia	93.77	26.84	93.86	24.83	100.42	31.54	101.45	36.96	101.32	31.22
India	66.93	0.00	69.03	0.00	68.88	0.00	64.49	0.00	70.10	0.00
Mexico	74.53	7.60	75.44	6.41	76.37	7.49	69.65	5.16	71.63	1.53
Pakistan	49.23	-17.70	53.20	-15.83	51.49	-17.39	48.62	-15.87	52.36	-17.74
Philippines	80.43	13.50	83.56	14.53	85.43	16.55	80.45	15.96	93.71	23.61
Thailand	50.57	-16.36	65.26	-3.77	71.41	2.53	71.66	7.17	70.36	0.26

Source: FAO.org

Annexure-VI

Export- import of sugar on financial year basis (2000-01 to 2011-12)

Financial year (April to March)	EXPORT		IMPORT	
	Quantity (Tonnes)	Value (Rs. in Crores)	Quantity (Tonnes)	Value (Rs. in Crores)
2000-01	338691	430.98	30404	31.11
2001-02	1456448	1728.29	26578	32.60
2002-03	1662370	1769.49	41430	32.83
2003-04	1200600	1216.59	74400	62.70
2004-05	108690	149.53	932740	976.18
2005-06	321204	569.11	558769	651.59
2006-07	1643403	3127.47	1052	3.49
2007-08	4684554	5412.16	496	2.24
2008-09	3331997	4448.74	386099	583.11
2009-10	44045	110.23	2424045	5961.24
2010-11	3249300	10352.27	1004100	2723.21
2011-12 (Up to September, 11)	2717300	8736.15	70454	209.64

Source: Cooperative Sugar, vol-44, No.4, Dec, 2012.

Annexure-VII

COUNTRY WISE EXPORT (TONNES) OF SUGAR FROM INDIA BY ISEC DURING 2001 TO 2010

S. No.	Country	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
1	Afghanistan	2100	-	2496	286	-	-	-	-	-	-	13883
2	Albania	-	-	-	-	-	-	3500	-	-	-	-
3	Bangladesh	12500	127737	80134	13168	-	-	2330	86531	-	-	309759
4	Belgium	10000	9000	10016	8200	-	9622	-	-	482	4723	-
5	China	-	-	-	-	-	-	1300	2650	-	-	5001
6	Croatia	-	-	-	-	-	-	3195	-	-	-	8390
7	Cyprus	-	-	-	-	-	360	-	-	264	-	-
8	Egypt	-	-	48883	-	-	-	-	-	-	-	11232
9	Eritrea	-	-	-	-	-	-	-	37500	-	-	16672
10	Ethiopia	-	-	-	-	-	-	2494	21000	-	-	-
11	Estonia	-	-	-	-	-	-	-	-	240	-	-
12	France	210	-	-	50	-	240	6024	-	208	-	-
13	Georgia	-	-	15800	-	-	-	-	-	-	-	3096
14	Germany	-	-	-	-	-	-	-	-	390	-	-
15	Greece	-	-	-	-	-	3312	-	9854	4394	-	-
16	Indonesia	-	16750	17600	-	-	-	-	26000	-	-	70192
17	Iran	-	-	-	-	-	-	260	35700	-	-	54801
18	Iraq	-	945	13125	-	-	-	4993	-	-	-	422
19	Italy	8547	1000	-	950	-	-	-	-	66	-	-
20	Korea	-	-	572	156	-	-	-	17000	-	-	-
21	Latvia	-	-	-	-	-	-	-	-	240	-	-
22	Malaysia	-	64524	21138	1534	-	-	-	44603	-	-	9709
23	Malta	-	-	-	-	-	-	-	-	-	220	-
24	Pakistan	-	-	-	-	-	335782	1248	-	-	-	53151
25	Philippines	11500	-	-	-	-	-	-	-	-	-	-
26	Portugal	10460	20217	10788	-	10079	10070	16350	-	-	-	-
27	Russia	-	-	-	-	-	-	625	-	-	-	1068
28	Rwanda	-	-	43	75	-	-	-	-	-	-	-
29	Saudi Arabia	-	-	-	-	-	-	650	11432	-	-	
30	Senegal	-	-	22	-	-	-	-	-	-	-	6840
31	Singapore	-	-	3900	468	468	-	-	-	-	-	8035
32	Somalia	-	-	-	-	-	-	2700	19052	-	-	135565
33	Spain	525	-	3050	-	-	-	-	-	254	223	-
34	Srilanka	750	44289	49462	15146	-	-	10485	21684	-	-	412649
35	Taiwan	-	-	-	-	-	-	260	-	-	-	-
36	Tanzania	-	-	430	-	-	-	4975	260	-	-	18568
37	Uganda	-	-	910	-	-	-	2030	-	-	-	4443
38	UAE	-	16887	260	-	-	-	164937	357419	-	-	260321
39	UK	-	-	-	-	-	-	-	-	-	-	-
40	USA	-	8140	8140	7957	-	11087	9900	-	-	-	2228
41	Yemen	-	-	36443	-	-	-	50778	23074	-	-	210309
42	Zambia	-	-	-	43	-	-	-	-	-	-	-
	Total	56592	309489	323212	48033	10547	370473	289034	713759	6538	5166	3016918

Annexure-VIII

Released and notified varieties of sugarcane

S. No.	Name of variety	Maturity group	Notification Number and date	Recommended area of cultivation
1	Cos 91230 (Raseeli)	Mid Season	S.O.821(E)13.09.2000	Punjab, Haryana, Rajasthan and Central Western Uttar Pradesh
2	Co Pant 90223	Midlate	S.O.821(E)13.09.2000	Uttar Pradesh, Harayana and Panjab
3	Uttara (Co87044)	Midlate	S.O.821(E)13.09.2000	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh, Tamil Nadu,Kerala
4	Bhavani (Co 86249)	Midlate	S.O.821(E)13.09.2000	Orissa, Coastal Andhra, Pradesh, Coastal Tamil Nadu, Pondicherry
5	Kalyani (Co 87025)	Midlate	S.O.821(E)13.09.2000	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh, Tamil Nadu, Kerala
6	Prabha (Co 85004)	Early	S.O.821(E)13.09.2000	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh, Tamil Nadu, Kerala
7	Dhanush (Co 91010)	Midlate	S.O.821(E)13.09.2000	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh, Tamil Nadu, Kerala
8	Nayana (Co 86032)	Midlate	S.O.821(E)13.09.2000	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh Tamil Nadu, Kerala
9	Bhima (Co 8371)	Midlate	S.O.821(E)13.09.2000	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh, Tamil Nadu, Kerala
10	Moti (87268)	Early	S.O.821(E)13.09.2000	Eastern UP, Bihar, West Bengal and North East regions of North Central and North East Zones
11	Sarayu (Co-87263)	Early	S.O.821(E)13.09.2000	Eastern UP, Bihar, West Bengal and North East regions of North Central and North East Zones
12	CoM 7714 (Krishna)/ CoM 88121	Midlate	S.O.821(E)13.09.2000	Chhattisgarh, Gujarat, Karnataka, Kerala, Maharashtra, interior Andhra Pradesh and plateau region of Tamil Nadu
13	Pramod (B.O.-128)	Midlate	S.O.92(E) 02.02.2001	Bihar, Eastern UP ,West Bengal
14	Haryana-92 (CoH-92201)	Medium	S.O.92(E) 02.02.2001	North West Zone
15	Gandak (Co-89029)	Early	S.O.1134(E) 15.11.2001	Eastern UP, Bihar, West Bengal and North East regions of North Central and North East Zones
16	Rasbhari (CoSE-95422)	Medium	S.O.1134(E) 15.11.2001	Uttar Pradesh , Bihar and West Bengal
17	Rajbhog (CoSE-92423)	Medium	S.O.1134(E) 15.11.2001	Uttar Pradesh , Bihar and West Bengal
18	CoS-767	Mid to late	S.O.1134(E) 15.11.2001	Haryana and Uttra Khad
19	Madhumathi	Midlate	S.O.937(E) 04.09.2002	Kerala

20	Rashmi (CoS-96234)	Early	S.O.642(E) 31.05.2004	Eastern UP, Bihar and West Bengal
21	Jalpari (CoS-96434)	Mid Late	S.O.642(E) 31.05.2004	Uttar Pradesh , Bihar and West Bengal
22	Rachana (CoS-95255)	Early	S.O.642(E) 31.05.2004	U.P. Punjab, Haryana and Rajasthan
23	Shyama (Co-94008)	Early	S.O.161(E) 04.02.2004	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh, Tamil Nadu, Kerala
24	Madhuri (Co-394164)	Medium	S.O.161(E) 04.02.2004	
25	Haryana Ganna-119 (CoH-119)	Medium	S.O.1566(E) 05.11.2005	Haryana
26	Haryana Ganna-110 (CoH-110)	Medium	S.O.1566(E) 05.11.2005	Haryana
27	CoC(sc)-22	Medium	S.O.1566(E) 25.08.2005	
28	Sweta (CoS-94270)	Medium	S.O.122(E) 02.02.2005	West Uttar Pradesh, Punjab, Uttaranchal, Haryana And Rajasthan
29	CoC(Sc)23 (COC-01-61)	Early	S.O.1572(E) 29.09.2006	Orissa, Andhra Pradesh, Pondicherry And Tamil Nadu
30	COSI(Sc)6	Early	S.O. 599(E) 25.04.2006	Tamil Nadu
31	CoG(sc)5	Medium	S.O. 599(E) 25.04.2006	Tamil Nadu
32	CoS-96268 (Mithas)	Early	S.O. 1178(E) 20.07.2007	Rajasthan, Uttranchal, Haryana, Punjab And Western Uttar Pradesh
33	CoS-96275 (Sweety)	Midlate	S.O. 1178(E) 20.07.2007	Haryana, Punjab, Rajasthan,Uttrakhand And Western and Central Uttar Pradesh
34	CoC-671 (Vasanth-1)	Early	S.O. 1178(E) 20.07.2007	Madhya Pradesh
35	Sarada (93A145)	Medium	S.O. 1178(E) 20.07.2007	Andhra Pradesh
36	Co Pant 97222	Medium	S.O. 122(E) 06.02.2007	Uttaranchal, Western and Central Uttar Pradesh, Haryana, Punjab and Rajasthan
37	Co-98014 (Karan-1)	Early	S.O. 122(E) 06.02.2007	Punjab, Haryana, Rajasthan, Uttarakhand, Western and Central UP
38	Co-99004 (Damodar)	Midlate	S.O. 122(E) 06.02.2007	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh, Tamil Nadu, Kerala
39	COJ-89		S.O. 1178(E) 20.07.2007	
40	CoLK 94184 (Birendra)	Medium	SO 2458 (E) 16.10.2008	U.P., Bihar and West Bengal
41	Co 0232	Early	SO 454 (E) 11.02.2009	Eastern UP, Bihar, West Bengal and North East regions of North Central and North East Zones
42	Co 0233	Midlate	SO 454 (E) 11.02.2009	Eastern UP, Bihar, West Bengal and North East regions of North Central and North East Zones
43	Sulabh(Co-2001-13)	Midlate	SO 454 (E) 11.02.2009	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh, Tamil Nadu, Kerala
44	Co-2001-15	Midlate	SO 454 (E) 11.02.2009	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh, Tamil Nadu, Kerala

45	Co-0238 (Karan-4)	Early	SO 454 (E) 11.02.2009	Punjab, Haryana, Rajasthan, Uttarakhand, Western and Central UP
46	Co-0118 (Karan-2)	Early	SO 449 (E) 11.02.2009	Punjab, Haryana, Rajasthan, Uttarakhand, Western and Central UP
47	C.O.-94012	Midlate	SO 449 (E) 11.02.2009	Maharashtra and Karnataka
48	Co-0239 (Karan-6)	Early	S.O. 2136(E) 31.08.2010	Punjab, Haryana, Rajasthan, Uttarakhand, Western and Central UP
49	Co-0124(Karan-5)	Midlate	SO 449 (E) 11.02.2009	Punjab, Haryana, Rajasthan, Uttarakhand, Western and Central UP
50	Co-0218	Midlate	S.O. 2136(E) 31.08.2010	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh, Tamil Nadu, Kerala
51	CoC(sc)24	Early	S.O. 2136(E) 31.08.2010	Tamil Nadu and Puducherry
52	Co-92005	Early	S.O. 283(E) 07.02.2011	Maharashtra
53	CoH-128 (Haryana Ganna128)	Medium	S.O. 456(E) 16.03.2012	Haryana, Punjab, Rajasthan, Uttaranchal
54	Co-05011(Karan-9)	Midlate	S.O. 1708(E) 26.07.2012	Punjab, Haryana, Rajasthan, Uttarakhand, Western and Central UP
55	Co.0237 (Karan-7)	Early	S.O. 2125(E) 10.09.2012	Punjab, Haryana, Rajasthan, Uttarakhand, Western and Central UP
56	Co.0403	Early	S.O. 2125(E) 10.09.2012	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh, Tamil Nadu, Kerala
57	Uttara		S.O. 952 (E) 10.04.2013	
58	Kanakmahalakshmi		S.O. 952 (E) 10.04.2013	
59	CoSe-01421 (Imarti)		S.O.2815 (E) 19.09.2013	
60	CO-05009 (Karan-10)	Early	S.O.2815 (E) 19.09.2013	Haryana, Punjab, Rajasthan, Uttaranchal and Western and Central Uttar Pradesh
61	Co-06030	Midlate	S.O.2815 (E) 19.09.2013	Orissa, coastal Andhra Pradesh and coastal Tamil Nadu
62	CO-06027	Midlate	S.O.2815 (E) 19.09.2013	Chhattisgarh, Gujarat, Karnataka, Kerala, Maharashtra, interior Andhra Pradesh and plateau region of Tamil Nadu

Annexure - IX

Varieties resistant to stresses, suitability to biotic stresses

List of varieties relatively resistant to different pests

Pest	Varieties
Shoot Borer	Co243, 281, 285, 312, 356, 421, 449, 453, 527, 617, 775, 853, 975, 1007, 1048, 1049, 6239, 6402, 6403, 6507, 6508, 6510, 6601, 6610, CoS 673, 729 BO 17, 34,70, and 99
Internode Borer	Co285, 453, 513, 617, 853, 915, 1007, 1287, 6806, and CoJ 46
Stalk borer	Co 395, 951, 1007, 1236, 6303, 7302, 7303, CoJ 46 and CP 43/47
Top borer	Co 205, 285, 331, 421, 453, 513, 617, 1007, 1111, 1158, 1330, 1349, 7224, CoJ 67 and CoL 9
Pyrilla	Co 313, 515, 545, 615, 612, 628, 738, CoL 22 and BO 3
Whitefly	Co 229, 312, 313, 385, 386, 513, and 617
Scale insect	Co 678, 1132, 6501, 6907, 7002, 7314,7411, 62174, 8008, 8009 8014
White grub	Co285, 453, 975, 1158, 1287, 6304, CoS 510 and CoJ 67

Annexure-X

Sugarcane varieties tolerant to drought, water logging and salinity Developed in India

S.No	Variety	Year of Release	State where recommended for cultivation	Reaction to abiotic stresses
1	Co-98014 (Karan-1)	2007	Punjab, Haryana, Rajasthan, Uttarakhand, Western and Central UP	Tolerant to drought
2	Co-99004 (Damodar)	2007	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh, Tamil Nadu, Kerala	Tolerant to drought
3	C.O.-94012	2009	Maharashtra and Karnataka	Tolerant to drought
4	Co-2001-13(Sulabh)	2009	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh, Tamil Nadu, Kerala	Tolerant to drought
5	Co-2001-15	2009	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh, Tamil Nadu, Kerala	Tolerant to drought
6	Co-0118 (Karan-2)	2009	Punjab, Haryana, Rajasthan, Uttarakhand, Western and Central UP	Tolerant to drought and water logging
7	Co 0232	2009	Eastern UP, Bihar, West Bengal and North East regions of North Central and North East Zones	Tolerant to water logging
8	Co 0233	2009	Eastern UP, Bihar, West Bengal and North East regions of North Central and North East Zones	Tolerant to water logging
9	Co-0238(Karan-4)	2009	Punjab, Haryana, Rajasthan, Uttarakhand, Western and Central UP	Tolerant to drought and water logging
10	Co-0239 (Karan-6)	2010	Punjab, Haryana, Rajasthan, Uttarakhand, Western and Central UP	Tolerant to drought and water logging
11	Co-0218	2010	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh, Tamil Nadu, Kerala	Tolerant to drought
12	Co.0237 (Karan-7)	2012	Punjab, Haryana, Rajasthan, Uttarakhand, Western and Central UP	Tolerant to drought and water logging
13	Co.0403	2012	Gujarat, Maharashtra, Madhya Pradesh, Karnataka, Interior Andhra Pradesh, Tamil Nadu, Kerala	Tolerant to drought

Salt stress	Co205, Co 285, Co 453, Co 6801, Co 62422, Co 1111, CoH 70, CoJ 1328, Co 449, Co 997 CoA 7602,Co 87263, Co 8347, Co 85004, Si 87241, Si 87231, So S86071, Co 85019, CoG 93076,CoG 95076, Co8208, CoC 671, Co8145, Co7717, CoS 767, Co93009, Co 93011, Co 93016, Co 93015, Co 93021, CoG 9307,

Source: Sugarcane Production Management and Agro industrial imperatives by Dr. S. Solomon et. al. Pg 293 & 439

Annexure- XI

State wise farmer's preferred / ruling varieties under cultivation

Andhra Pradesh

Variety	2008-09	2009-10	2010-11
Early varieties: Co.6907, 84A125, 81A99, 83A30, 85A261, 87A298, Co.8014, 86V96, 91V83, and others	7000	62000	92000
Mid-late Varieties: COA7607, CO8021, COT.8201, Co7805, COV92102 (83V15), 83V288 and others	40000	34000	51000
Late varieties: Co.7219, CoR8001, 87A380, Co7706, and others	29000	24000	37000

Bihar

Sl.No.	Variety	2008-09	2009-10	2010-11
1	BO.99	6281	4309	3929
2	BO-120	2546	2818	2510
3	BO-130	4384	6874	6086
4	BO-139	100	402	2217
5	BO-141	0	120	188
6	BO-144	0	0	57
7	BO-145	149	117	415
8	COP-9301	7629	12824	19285
9	COSE-95436	267	421	1228
10	COSE-98231	1619	2449	4463
11	COS-8436	13525	6604	6820
12	COS-88230	2699	3297	5961
13	COS-95255	9293	8457	6961
14	COS-96268	759	1186	2543
15	COS-89029	878	1623	181
16	COS-95130	82	103	225
17	BO-102	5906	399	360
18	BO-138	232	83	727
19	COS-687	0	83	0
20	BO-76	0	322	2426
21	BO-91	76379	57476	62908
22	BO-110	16937	19331	22940
23	BO-128	4554	2364	2586
24	BO-136	2064	2319	3408
25	BO-137	9107	12010	13665
26	BO-147	954	3030	7343
27	BO-108	150	98	515

28	BO-109	539	269	469
29	COP-9206	12408	14382	18014
30	COP-9302	4150	5452	6864
31	COS-767	6581	4051	8152
32	COSE-95422	21456	28379	28123
33	COSE-95427	0	13	2433
34	COS-8432	4921	3428	7084
35	COLK-8102	2506	1286	1922
36	COSE-92423	8806	6700	14979
37	COP-9702	150	185	245
38	UP-9530	7267	6550	11152
39	COSE-96436	257	710	1261
40	CO-97264	189	139	588
41	COS-91269	808	0	2657
42	CO-1148	135	0	251
43	CO-1158	1866	0	341
44	COJ-83	66	0	35
45	HR-90	0	136	0
46	COJ-88	146	612	1268
47	COJ-85	534	990	1452
48	Others	25377	28986	24848
	Total	**264656**	**251387**	**312085**

Gujarat

S.No.	**Variety**	**2008-09**	**2009-10**
1	Coc.671	673	2638
2	Co.86002	17837	17853
3	Co.97009	7560	3660
4	Co.86032	42812	53622
5	Cosi.95071	16643	9380
6	CoN.91132	1903	0
7	Co.86249	32064	27025
8	Co 6304	164	318
9	CoN 05071	12210	21955
10	CoN 05072	1376	1941
11	Co 99004	0	904
12	CoM 0265	0	371
	others	4928	11448
	Total:	**221300**	**153979**

Haryana

S. no.	Variety	2006-07	2007-08	2008-09
1	**Early Group**			
2	Coj-64	11210	9958	
3	Coj-85	2050	3524	5605
4	Coj-83	1500	1454	
5	Cos-92	350	311	290
6	Cos-56	2340	2078	404
7	Others	940	4507	7509
8	**Mid Group**			
9	Co-7717	5090	4525	836
10	Coh-99	990	4434	1089
11	Cos-8436	61565	56548	35371
12	Cos-88230	11705	11375	
	Others	9650	14735	12290
	Late Group			
13	Cos-767	20190	19704	5463
14	Co-1148	895	797	353
15	Coh-110	325	3005	2410
	Others	3010	3046	2381
	Total	**131810**	**140000**	**74001**

Karnataka

Variety		2008-09	2009-10	2010-11
	Early			
1	**COC 671**	60000	45000	64000
2	**CO 92020**	40000	10000	12000
3	**CO 94012**	20000	40000	10000
	Mid			
4	**CO 8014**	30000	10000	10000
5	**CO 86032**	40000	131000	195000
6	**CO 62175**	25000	30000	50000
7	**CO 8371**	22000	5000	7000
	Late			
8	**CO 740**	25000	15000	15000
9	**CO 8011**	15000	42500	55000
10	**CO 419**	4000	8021	4000
	Total	**281000**	**336521**	**422000**

Maharashtra

S.No.	Variety	2008-09	2009-10	2010-11
1	**CO 740**	5500	1000	1000
2	**CO 7219**	2100	500	2000
3	**COM 7125**	100	1000	1000
4	**CO 7527**	3900	2000	2000
5	**CO 8014**	8900	4000	7000
6	**COC 671**	241400	178000	191000
7	**COM 88121**	900	100	0
8	**CO 86032**	404600	438000	547000
9	**CO 419**	2800	5000	4000
10	**CO 8011**	18300	4000	3000
11	**CO 8021**	1600	2000	0
12	**CO 94012**	38700	19000	20000
13	**VSI 434**	2100	1000	3000
14	**COM 265**	4600	50000	129000
15	**Co VSI 9805**		1000	4000
16	**CO 92005**			41000
	Others	32900	29000	10000
	Total	**768000**	**735600**	**965000**

Odisha

Sl.No.	Variety	2008-09	2009-10	2010-11
1	Co.62175	13.28	11.74	11.5
2	Co.86032	3.04	3.1	3.8
3	Co.A.89085	4.55	4.65	4.7
4	Co.7805	1.14	1.1	1.12
5	Co.8021	1.9	1.95	2.25
6	Co.87A298	2.26	2.3	3.15
7	Co.86V96	2.28	2.38	2.56
8	Co.95A145	1.9	1.95	1.6
9	Others	7.59	7.69	9.15
	Total	**37.94**	**36.86**	**39.83**

Punjab

S.No.	Variety	2008-09	2009-10	2010-11
1	**Early varieties**			
2	CoJ 64	1111	903	881
3	CoJ83	1558	1317	1429
4	CoJ85	28905	21940	24527
	Total	**31574**	**24160**	**26837**
	Mid Varieties			
5	CoJ 88	7808	6717	1305
6	CoS 8436	10017	9812	9888
7	COH-119	7273	6320	5901
	Total	**25098**	**22849**	**17094**
	Late Varieties			
8	Co89	85	70	46
9	Co1148	317	133	119
10	Co-89003	11009	8551	9236
	Total	**11411**	**8754**	**9401**
	Others	**12997**	**4255**	**10681**
	Grand Total	**81080**	**60018**	**70014**

Tamilnadu

Sl. No.	Varities	2007-08	2008-09	2009-10
1	Co..94012	290298	268656	279551
2	Co.92012	38942	5404	785
3	Co.94010	10621	13124	12564
4	CoC.24	7080	9264	9423
5	Others	4425	4632	5497
	Total	**351366**	**301080**	**307820**

Uttar Pradesh

Sl.No.	Varieties	2008-09	2009-10	2010-11
	Early Varieties			
1	Co.J. 64	43090	24979	18972
2	Co.S. 8436	121976	86869	78032
3	Co.S. 88230	91014	64059	57804
4	Co.S. 95255	18471	13771	12164
5	Co.S. 96268	11107	8675	8183
6	Co.S. 98231	22972	27245	32301
7	**Others**	**10197**	11957	**14664**
	Total	**318827**	237555	**222120**
	Mid and Late varieties			
	Suitable			
1	U.P. 39	17973	14657	15087
2	Co.S. 767	701184	612435	705731
3	Co.S. 8432	106205	97025	96824
4	Co.Pt. 90223	1718	20292	21639
5	Co.S. 92423	497093	469690	565205
6	Co.S. 94257	6625	6290	7737
7	Co.Se. 95422	45041	43379	51308
8	Co.S. 97264	89718	75247	103017
9	Others	0	23841	47249
	Total	**1465557**	**1362856**	**1613797**
	Water Logging			
1	U.P. 9530	12157	9174	12285
2	Co.S. 96436	2222	1662	1782
3	**Total**	**14379**	10836	**14067**
	Total Suitable	**1170026**	**1373692**	**1627864**
	Unsuitable			
1	Co.Lk. 8102	34467	38619	26294
2	Co.S. 90260	150	4	199
3	Co.Se.96232	6	123	202
4	Co.Se. 96234	0	0	0
5	U.P. 22	0	0	0
6	B.O. 91	5577	2250	2832
7	B.O.128	118	0	24
8	Co.S. 1148	4767	1223	1523
9	Co.S.7918	571	436	1255
10	Co.S. 91269	133343	84167	107329
11	**Others**	**162009**	**49768**	**111755**
	Total (Unsuitable)	**341008**	**176608**	**251413**
	Total (Midlate)	**1820944**	**1550300**	**1879277**
	Grand Total	**2139771**	**1787855**	**2101397**

Uttarakhand

S.No.	Early variety	2008-09	2009-10	2010-11
1	Cos.8436	15005	12785	16002
2	Cos.88230	15174	14285	12235
3	Cos.96258	66	56	25
4	Cos.96268	229	305	244
5	Coj.64	495	159	144
6	Cos.98247	5	535	805
7	CoPant-94211	589	39	12
8	Coj-85	14	347	362
	Mid Late variety			
9	Cos.767	41334	39250	45437
10	Co.Pant.84212	4681	3562	3286
11	Co.Pant.90223	3307	2270	2516
12	CoS-97264	7791	5116	6488
13	CoSe-92423	11153	8201	9313
14	Cos.8432	3181	2794	3245
15	Cos.94257	554	382	462
16	Co.Pant.96219	125	43	9
17	UP-9530	99	82	74
18	CoSe-96436	159	99	125
19	CoPant-97222	471	938	1509
20	CoPant-99214		72	343
20	CoJ 88	287	120	98
21	Others	2258	1578	2126
	Total	**107251**	**93018**	**104860**

West Bengal

Sl.No.	Variety	2008-09	2009-10	2010-11
1	B.O. -91	7025	5500	6000
2	COB-94104 (Madhuri)	3510	2750	3000
3	CO-62033	1755	1375	1500
4	Local	5267	4125	4517
	Total	**17557**	**13750**	**15017**

Annexure-XII

SUGARCANE, SUGAR AND MOLASSES PRODUCTION AT A GLANCE

STATEMENT SHOWING AREA, PRODUCTION AND YIELD OF SUGARCANE, FACTORIES IN OPERATION, DURATION, CAPACITY, CANE CRUSHED, SUGAR AND MOLASSES PRODUCTION & THEIR RECOVERY % CANE

Year	Area under Sugarcane (000 ha.)	Production of sugarcane (000 tones)	Yield of cane per ha. (tones)	No. of factories in operation	Average duration days	Average capacity (tones/ day)	Total cane crushed (000 tonnes)	Total Sugar produced (000 tonnes)	Recovery of sugar (% cane)	Molasses production (000 tonnes)	Molasses (% cane)
1930-31	1176	36354	30.90	29	-	-	1339	120	8.96	-	-
1940-41	1617	51978	32.10	148	113	750	11492	1113	9.70	431	3.76
1950-51	1707	54823	32.10	139	101	882	11348	1100	9.99	387	3.60
1960-61	2415	110001	45.50	174	166	1172	31021	3021	9.74	1210	3.99
1970-71	2615	126368	48.30	215	139	1394	38205	3740	9.79	1611	4.22
1980-81	2667	154248	57.80	315	104	1718	51584	5150	9.98	2126	4.12
1990-91	3686	241045	65.40	385	166	2088	122338	12047	9.84	5454	4.45
2000-01	4316	295956	68.60	436	138	3203	176660	18511	10.48	7820	4.43
2001-02	4411	297208	67.40	434	138	3285	180346	18528	10.27	8073	4.48
2002-03	4520	287383	63.60	453	140	3343	194365	20145	10.36	8879	4.57
2003-04	3938	233862	59.40	422	99	3493	132511	13546	10.22	5905	4.46
2004-05	3662	237088	64.80	400	97	3508	124772	12690	10.17	5513	4.42
2005-06	4201	281172	66.90	455	125	3619	188672	19267	10.21	8549	4.53
2006-07	5151	355520	69.00	504	173	3561	279295	28367	10.16	13111	4.69
2007-08	5055	348188	68.90	516	149	3586	249906	26357	10.55	11313	4.53
2008-09	4415	285029	64.60	489	87	3751	144983	14539	10.03	6546	4.51
2009-10	4175	292302	70.00	490	109	3846	185548	18912	10.19	8400	4.53
2010-11	4886	342382	70.10	527	135	3744	239807	24394	10.17	10970	4.57
2011-12*	5085	357667	70.30	529	137	3877	256975	26343	10.25	11824	4.60

* *Provisional*　　Source: Cooperative Sugar, vol-44, No.4, Dec, 2012.

Annexure-XIII

STATEMENT SHOWING FACTORIES IN OPERATION, OPENING STOCKS, PRODUCTION, IMPORTS, CONSUMPTION AND EXPORTS OF SUGAR (Lakh Tonnes) DURING LAST 10 YEARS

Season	No. of factories in operation	Carry over of sugar in the beginning of the season	Production during the year	Imports	Total available supply	Consumption	Exports	Closing stock of Sugar
2000-01	436	94.34	185.11	-	279.45	162.00	9.87	107.58
2001-02	434	107.58	185.28	-	292.86	167.81	10.94	114.14
2002-03	453	114.14	201.45	0.41	316.00	183.84	15.00	117.16
2003-04	422	117.16	135.46	4.00	256.62	172.85	2.94	80.83
2004-05	400	80.83	126.90	21.38	229.11	185.00	0.04	44.07
2005-06	455	44.07	192.67	-	236.74	189.45	11.07	36.22
2006-07	504	36.22	283.67	-	319.89	201.60	17.28	101.01
2007-08	516	101.01	263.57	-	364.58	220.00	49.57	95.01
2008-09	489	95.01	145.39	24.03	264.43	230.00	2.17	32.26
2009-10	490	32.26	189.12	40.80	262.18	210.00	2.35	49.83
2010-11	527	49.83	243.94	-	297.77	207.36	26.00	60.41
2011-12*	529	60.41	263.43	-	323.84	220.00	33.90	69.94

** Provisional*

Source: Cooperative Sugar, vol-44, No.4, Dec, 2012.

Annexure-XIV

UTILIZATION OF SUGARCANE FOR DIFFERENT PURPOSES

Year	Production of Sugarcane (000 tonnes)	Cane used for (000 tonnes)			Percentage of sugarcane production utilized for		
		Production of white sugar	Seed, feed & chewing etc.	Gur and Khandsari	Production of white sugar	Seed, feed & chewing etc.	Gur and Khandsari
2000-01	295956	176660	33930	85366	59.70	11.50	28.80
2001-02	297208	180346	34724	82138	60.70	11.70	27.60
2002-03	287383	194365	33524	59494	67.60	11.70	20.70
2003-04	233862	132511	27830	73521	56.70	11.90	31.40
2004-05	237088	124772	28213	84103	52.60	11.90	35.50
2005-06	281172	188672	33459	59041	67.10	11.90	21.00
2006-07	355520	279295	42307	33918	78.60	11.90	09.50
2007-08	348188	249906	40525	50126	71.80	11.60	16.60
2008-09	285029	144983	33833	106213	50.90	11.90	37.20
2009-10	292302	185548	34784	71970	63.50	11.90	24.60
2010-11	342382	239807	40743	61832	70.00	11.90	18.10
2011-12*	357667	256975	42527	58165	71.90	11.90	16.20

** Provisional*

Source: Cooperative Sugar, vol-44, No.4, Dec, 2012.

Annexure-XV

STATE WISE UTILIZATION (%) OF SUGARCANE FOR SUGAR PRODUCTION IN MAJOR STATES

S.	State	2006-07	2007-08	2008-09	2009-10	2010-11	2011-12*
1	**A.P.**	79.86	65.18	38.59	47.38	68.95	69.27
2	**Bihar**	87.37	89.13	54.46	54.13	32.44	39.44
3	**Gujarat**	85.67	75.97	63.89	91.09	89.82	66.53
4	**Haryana**	69.89	68.45	31.40	49.63	71.93	70.03
5	**Karnataka**	87.89	109.08	92.78	78.76	85.14	89.55
6	**M.P.**	68.82	57.30	14.26	33.65	63.74	61.23
7	**Maharashtra**	101.67	94.47	70.16	95.68	97.96	94.14
8	**Orissa**	49.06	62.30	29.26	51.23	57.49	82.61
9	**Punjab**	84.57	85.97	41.32	57.08	82.33	91.46
10	**Tamilnadu**	66.75	59.90	47.20	48.17	59.30	64.80
11	**U.P.**	66.81	59.95	37.86	48.43	53.41	59.66
12	**Uttarakhand**	91.92	53.47	31.50	54.33	49.79	30.99
	All India	**78.56**	**73.38**	**49.20**	**63.48**	**70.04**	**71.85**

*** Provisional**

Source: Cooperative Sugar, vol-44, No.4, Dec, 2012.

Annexure-XVI

STATE WISE CANE CRUSHED (in '000 tonnes) BY SUGAR FACTORIES IN INDIA

S. No.	State	2002-03	2003-04	2004-05	2005-06	2006-07	2007-08	2008-09	2009-10	2010-11	2011-12*
1.	AP	11980	8597	9217	12303	17323	13201	5993	5547	10317	11588
2	Bihar	4537	2932	2650	4455	5204	3639	2370	2724	4141	4761
3.	Chhattisgarh	39	175	102	185	310	424	150	110	267	435
4	Goa	139	112	89	121	201	148	108	100	146	116
5.	Gujarat	11828	9753	7405	10787	13390	12801	306	11295	12360	9432
6.	Haryana	6276	5560	3938	4187	6695	6055	9445	2648	4346	5430
7.	Karnataka	17303	10942	10283	17953	25197	26685	2528	23977	33765	34753
8.	MP	717	917	732	959	1931	1822	625	853	1700	1639
9	Maharashtra	53441	29077	19456	44578	79884	76144	40023	61390	80223	77063
10.	Orissa	426	445	466	442	625	666	327	251	519	731
11.	Pudducherry	354	206	185	371	715	576	166	224	546	720
12.	Punjab	6035	4006	3220	3676	5091	5760	2603	2112	3433	4271
13.	Rajasthan	24	99	52	83	79	84	42	48	49	29
14.	Tamilnadu	16645	9282	11492	23185	27452	22970	16606	14328	20310	25455
15.	Uttar Pradesh	59271	46352	51472	60809	89494	74739	45482	56734	64381	76855
16.	Uttarakhand	5238	3973	3955	4519	5608	4110	2421	3174	3235	3641
17.	West Bengal	91	83	58	59	96	72	29	29	69	56
	All India	**194365**	**132511**	**124772**	**188672**	**279295**	**249906**	**144983**	**185548**	**239807**	**256975**

* Provisional

Source: Cooperative Sugar, vol-44, No.4, Dec, 2012.

Annexure-XVII

STATEMENT SHOWING THE MINIMUM STATUTORY PRICE/ FAIR & REMUNERATIVE PRICE OF SUGARCANE FIXED BY THE GOVERNMENT

Year	Minimum Statutory price of Sugarcane (Rs. per quintal)	Linked to basic sugar recovery % cane	Premium on every 0.1% increase in sugar recovery % cane (Rs. per quintal)	Range of Minimum Sugarcane Price on the basis of Col. 1, 2 & 3 (Rs. per quintal)
2000- 01	59.50	8.50	0.70	59.50 to 96.60
2001- 02	62.05	8.50	0.73	62.05 to 100.74
2002- 03	69.50	8.50	0.82	69.50 to 113.78
2003- 04	73.00	8.50	0.85	73.00 to 118.90
2004- 05	74.50	8.50	0.88	74.50 to 110.58
2005- 06	79.50	9.00	0.88	79.50 to 112.94
2006- 07	80.25	9.00	0.90	80.26 to 119.85
2007- 08	81.18	9.00	0.90	81.18 to 118.98
2008- 09	81.18	9.00	0.90	81.18 to 123.48
2009- 10	107.76 (SMP)	9.50	1.13	NA
	129.84 (FRP)	9.50	1.37	129.84 to 179.16
2010- 11	139.12 (FRP)	9.50	1.46	139.12 to 197.52
2011-12	145.00 (FRP)	9.50	1.53	145.00 to 203.14
2012-13	170.00 (FRP)	9.50	1.79	NA

Source: Cooperative Sugar, vol-44, No.4, Dec, 2012.

Annexure-XVIII

PER CAPITA CONSUMPTION OF SUGAR, GUR & KHANDSARI

Year	Population in Million (As on 1st March)	Consumption (lakh tones)		Per capita consumption (kg/annum)		Total per capita consumption ofsugar, gur & khandsari (kg/ annum)
		Sugar	Gur & Khandsari	Sugar	Gur & Khandsari	
1960-61	439	21.13	66.87	4.8	15.2	20.0
1970-71	546	40.25	74.37	7.4	13.6	21.0
1980-81	684	49.80	85.22	7.3	12.5	19.8
1990-91	846	107.15	90.71	12.7	10.7	23.4
2000-01	1029	162.00	86.09	15.7	8.4	24.1
2001-02	1043	167.81	83.11	16.1	8.0	24.1
2002-03	1060	183.84	56.94	17.3	5.4	22.7
2003-04	1077	172.85	71.46	16.0	6.6	22.6
2004-05	1093	185.00	81.75	16.9	7.5	24.4
2005-06	1106	189.45	57.39	17.1	5.2	22.3
2006-07	1122	201.60	33.38	18.0	3.0	21.0
2007-08	1138	220.00	50.93	19.3	4.5	23.8
2008-09	1154	230.00	107.92	19.9	9.3	29.2
2009-10	1170	210.00	73.12	17.9	6.2	24.1
2010-11	1186	207.36	59.94	17.5	5.1	22.6
2011-12*	1202	220.00	56.71	18.3	4.7	23.0

* Provisional

Source: Cooperative Sugar, vol-44, No.4, Dec, 2012

Annexure-XIX

Temperature requirement for different growth stages of sugarcane

S.No.	Critical Stages of sugarcane	Maximum temperature (^oC)	Minimum temperature (^oC)	Relative Humidity (%)
1.	Germination	32.0	20.0	-
2.	Tillering	35.0	18.0	-
3.	Grand Growth	30.0	14.0	80-85
4.	Ripening	30.0	20.0	50-55

Source: Sugarcane Production technology in India by Dr. R. S. Verma

Annexure-XX

Important growing / Planting and harvesting time of sugarcane in different States

Sl. No.	State	Time of sowing		Harvesting time
	Tropical Regions			
1	Andhra Pradesh	*Adsali*	Aug.-Sept	March-April
		Eksali	Jan-Mar	December-April
2	Assam	*Eksali*	Jan-Mar	November-March
3	Bihar	*Eksali*	-Oct -Feb-Mar	November-February Nov. – April
4	Gujarat	*Eksali*	-Jan-Feb -Oct-Nov	Nov.-Dec. Nov.-Dec.
5	Haryana	*Eksali*	-Oct -Feb-Mar	Nov. – Dec. Dec. – April
6	Karnataka	*Adsali*	Jul-Aug	Dec. – Jan.
		Eksali	-Oct-Nov -Jan-Feb	April-May Jan. – March
7	Kerala	*Eksali*	Oct-Dec	August – March
8	Madhya Pradesh	*Eksali*	-Oct-Nov -Jan-Feb	Dec. – April Jan- Feb.
9	Maharashtra	*Adsali*	June-July	Nov. – Dec.
		Eksali	-Oct-Nov -Jan-Feb	Jan. – Feb. Feb.-March
10	Orissa	*Eksali*	Jan-Mar	Dec.- May
11	Punjab	*Eksali*	Feb-Mar	Nov.-March
12	Rajasthan	*Eksali*	-Oct -Feb-Mar	 Nov.-March
13	Tamil Nadu	*Eksali*	-July-Sept -Feb to March & May -Jul-Sept	July-Sept. -Feb to March & May -Jul-Sept
14	Uttar Pradesh	*Eksali*	-End Sept-Oct -Feb-Apr	December Dec. – May
15	West Bengal	*Eksali*	Feb-Apr	Feb-Apr

Annexure-XXI

EXTENT OF LOSSES DUE TO DIFFERENT INSECT & PESTS IN INDIA

S. No.	Name of Pest	% reduction in cane yield	% reduction in sugar recovery
1.	Early shoot borer	22 to 33	2 CCS
2.	Internode borer	34.88	1.7-3.07
3.	Top shoot borer	21-37	0.2-4.1
4.	Stalk borer	Up to 33	1.7-3.07
5.	Gurdaspur borer	5-15	0.1-0.8
6.	Rood borer	35.00	0.3-2.90
7.	Scale insect	32.60	1.5-2.5
8.	Mealy bug	poor germination	brix loss 16.20
9.	Black bug	upto 35	0.1-2.8
10.	Pyrilla	31.60	2.0-3.0
11.	Arboridia sp.	14.70	1.0-1.5
12.	White Fly	86.00	1.4-1.8
13.	White grub(H)	80	5.0-6.0
14.	Whiter grub(L)	100	complete drying
15.	Termite	33	4.5
16.	Rodents	22.27	-
17.	Sugarcane woolly aphid	7 to 39	1.2-3.43

Annexure-XXII

Management schedule for insect pests of sugarcane for Sub-tropical India.

Crop stage	Pests covered	Control measures
Selection of crop for seed	Borers and scale insect	Seed should not be taken from fields having more incidence
Seed selection	Borers and scale insect	Setts showing borer damage should be discarded
Seed treatment	Scale insect	Dipping the setts in 0.1 % malathion I 0.08 % dimethoate for 15 minutes
At planting time	Termites, shoot borer, root borer	Application of chlorpyriphos or endosulfan at 1.0 Kg a. i. per ha over cane setts.
April- May	Shoot borers Thrips	Release S. *inference* @ 125 gravid females per ha. GV spraying 10^7- 10^8 IB per ml shoot borer Spraying with dimethoate 0.04%
June- July	Top borer	Soil application of 3 G carbofuran @ 1 Kg a. i. or 10 G *phorate* @ 3 Kg a. i. per ha against third brood of the pest in 2nd week of June in Eastern U. P. and Bihar, in 4th week of June 1st week of July in West U. P., Haryana and Punjab Release of laboratory bred *Isotima iavensis*
July August	White grubs	Hand collection of adult beetles and grubs. Soil treatment with *quinalphos* 5 G at 2.5 kg. a. i. per ha coinciding with rush of adult emergence.
July-September	Gurudaspur borer, Plassey Borer Pyrilla	Mechanical control of infested plants having gregarious Larval stage as large campaign. Redistribution and Colonization of *E. melanoleuca* cocoons or eggs masses Foliar spray of M. *anisopliae* @ 107 spores 1ml Release of Pyrilla adults seeded with M. *anisopliae* @ 250 adults Iha.
July-Oct.	Stalk borer, internode borer, Gurudaspur borer, root borer	Release of *T. chilonis* at 50,000 per ha at 10 days interval
October – November	Stalk borer	Detrashing of dried leaves 2 times at 30 days interval Removal of late shoots at 15 days intervals till harvest Spraying cane stalk with monocrotophos at 0.75 Kg a. i. per ha following detrashing.
November-December	Stalk borer Black Bug	*Beauria bassiana* spraying @ 10^{7} spores per ml to check the carry over borer population Release of adults seeded with spores of *Beauria bassiana* @ 5000 per ha to check the carry over population
Harvesting	For pests, in general (scale insect, mealy bug, black bug and pyrilla)	Deep harvesting Removal of water shoots or late shoots Trash burning

Annexure- XXIII

Management schedule for insect pests of sugarcane for Tropical India

S.No	Crop stage	Pests covered	Control practices
1.	Land preparation	White grub Termite Sugarcane woolly aphid (SWA)	Expose the grub stages by deep ploughings for predation. Apply 1 kg. of 2 % methyl parathion dust in 1 cart load of FYM or compost. Destroy the termitoria present on the bunds and nearer to the field Paired or wider row planting
2.	Selection of seed crop	Scale insect, mealy bug, white fly, borers, SWA	Seed should be free from the pest infestation and seed plots must be selected by concerned Agri. officers.
3.	Seed selection	Scale insect, mealy bug, white fly, borers	Sugarcane setts damaged by borers, scale insects etc. should be discarded. The discarded setts, leaves left after seed preparation should be buried.
4.	Sett treatment	Scale insect, mealy bug, SWA	Dipping of setts in 0.15% *malathion* or 1- 0.08% *dimethoate* for 10-15 minutes.
5.	At planting(January)	Termite, shoot borer, root borer	Soil application of 6 G *lindane* or 4:4 G *sevidol* @1 kg. a.i. per ha.or *Caldan* 4 G @ 0.5 kg. a.i. per ha. Drenching of 20 EC *chlorpyriphos* @ 1 kg. a.i. in 1000 lit. of *water per ha.* or soil application of 5 G *quinalphos* @ 1.5 kg. *a.i.per* ha.
6.	21 days after planting (January)	Shoot borer - - Spittle bug	Release *Trichogramma chilonis* @ 3 to 5 lakh parasitized eggs1ha. at 15 days intervals in suitable installments. Placement of pheromone sleeve traps @ *25 per ha.* for *C. infuscatellus* control & destroy male adults Spraying of 0.08% dimethoate
7.	45 days after planting (February)	Shoot borer White grub SWA Root borer	Small earthing up followed by trash mulching. Do not undertake maize, sorghum as intercrops. Prefer coriander, garlic, and onion as intercrops. Prefer paired row system for planting. Collect and destroy the grub stages during weeding. Needbase application of insecticide. Drenching of 20 EC *chlorpyriphos* @ 1 kg. a.i. per ha in 1000 lit. of water per ha. or soil application of 5 G *quinalphos* @ 1.5 kg. a.i per ha.
8.	60 days after planting(March)	White fly Leafhopper Pyrilla White grub (L) Scale insect and mealy bug, SWA	Remove 2-3 leaves containing pest stages. Spray 0.08% *monocrotophos* or DDVP with addition of 2.5 % N in spray solution or spray *neemark* @ 5 lit per ha. Release 1000 viable cocoons of *Epiricania* parasites per ha. Apply 10 G *phorate* @ 2.5 kg. a.i. per ha. Apply 10 G *phorate* @ 2.5 kg. a.i. per ha. Detrash the lower dry leaves and spray 0.08 %

		Rodents	*dimethoate*. Apply bromadiolone cake 0.005 % in rodent burrows or bait stations continuously for two days.
9.	90-120 days after planting, (April – May)	White grub (H) Internodes borer Topshoot borer White fly	Collection and destruction of beetles from neem trees during night time immediately after first heavy showers. Release *T. chilonis* @ 3 to 5 lakh parasitized eggs per ha. at 15 days intervals in suitable installments Destroy the egg masses and remove the affected canes along with pest stages. Avoid excess use of N fertilizers before earthing up.
10.	150 – 180 days after planting (June-July)	White grub (H) Pyrilla SWA Grasshopper	Collection and destruction of beetles from neem trees during night time immediately after first heavy showers. Soil application of 10 G *phorate* @ 2.5 kg. per ha or 2 % methyl parathion dust @ 2 kg. a. i. ha. Release 1000 viable cocoons of *Epiricania* parasites per ha. Spot spraying of bio-pesticide like *Verticillium* Dusting of 2 % methyl parathion dust @ 40 kg. per ha in sugarcane and on bunds.
11.	210 – 240 days after planting (August –September)	White fly Pyrilla SWA White grub (L) Army worm Leafhopper Rodents Nematodes Snail	Remove 2-3 leaves containing egg and pupal stages. Spray 0.08 % DDVP or monocrotophos with addition of 2.5 % N in spray solution or spray neemark @ 5 lit1ha. or release 1000 adults of *Chrysoperla carnae* predator per ha. Release 1000 viable cocoons of *Epiricania* parasites per ha. Augmentation of predators like Dipha, *Micromus* and *Syrphid* fly @ 1000 larvae or cocoons per ha. Collection and destruction of adults from sugarcane. Apply 10 G phorate @ 2.5 kg. a.i. per ha. Collection and destruction of larvae during rainy season. Remove 2-3 leaves containing pest stages. Spray 0.08 % DDVP or monocrotophos. Apply bromadiolone cake 0.005 % in rodent burrows or bait stations continuously for two days Soil application of 3 G carbofuran @ per kg. a.i. per ha. Cleaning of bunds and fields, collection and destruction of snails, dusting of lime on snails. Metaldehyde and iron_phosphate baiting.
12.	240 days onwards till harvest.	Internode borer Pyrilla	Release Trichogramma chilonis @ 3 to 5 lakh parasitized eggs per ha. at 15 days interval in suitable installments and use pheromone traps.

			Release 1000 viable cocoons of Epiricania parasites per ha.
13.	At harvest	Root borer	Harvesting at ground level to destroy the pest stages. Destroy stray of plants
14.	Post harvest	White fly Root borer White grub Snail SWA	In heavily affected fields burn the trash after harvest. Do not keep the ratoons in heavily affected fields. Use rotavator and subsoiler to destroy the pest stages Proper crop rotation

Annexure-XXIV

Important Diseases of Sugarcane and Their Management

Sl. No.	Name of disease	Causal agent	Symptoms	Disease management.
1	Red rot	*Glomerella tucumanensis*	The spindle leaves (3rd14th)) display drying. At a later stage, stalks become discoloured and hollow. Acervuli (black fruiting bodies) develop on rind and nodes. After splitting open the diseased stalk, a sour smell emanates. The internal tissues are reddened with intermingled transverse white spots. In advanced stage of the disease, the colour becomes earthy brown with pith cavity in the centre showing white cottony hyphae and sometimes fruiting bodies of fungus (acervuli). In rainy season, the disease spreads so fast that whole crop dries and not a single millable cane is obtained.	Resistant or moderately resistant varieties should be used. Any sett showing reddening at the cut ends or at the nodal region should be discarded. Healthy seed should be planted. Such seed must be produced from crop raised from heat treatment *of* seed canes in moist hot air at 54°C for 2.5 hour at 99% humidity. As soon as disease' is noticed, the affected clump along with root system should be uprooted and burnt. Bunding of affected field should be done to avoid movement of rain or floodwater. Ratooning *of* diseased crop should be avoided. Diseased crop should be harvested as early as possible. Crop rotation should be followed in affected fields.

2.	Smut	*Ustilago* scitaminea	The new sprouts are lean and lanky ,profuse in number and the growing point projects out a long black whip covered with black spores. Affected plants have slender and thin canes with erect and pointed leaves. Such plants can be easily located before the production of smut whip.	Resistant or moderately resistant varieties should be used. Healthy seed (as mentioned under red rot) should be planted. Pre-treatment of seed pieces by dipping in 2.5% organomercurial fungicide helps reduce the incidence. Removal of affected clump showing smut whip during tillering phase effectively reduced the disease incidence.
3.	Wilt	*Cephalosporium sacchari*	Wilt symptoms usually appear after monsoon. Infected clumps, individually or collectively, show stunting and yellowing of top leaves. In severe cases, whole clump dries, cane becomes hollow and lighter in weight. Red discoloration in internodes is more intense towards nodes which do not emit specific odour.	Healthy seed (as mentioned under red rot) should be planted. Seed setts may be treated with *organomercurial* fungicide before planting. Crop rotation should be followed in affected field.
4.	Grassy shoot	Mycoplasma like organism (MLO)	A large number lean and lanky, pale sprouts in the clump appear like a 'bunchy grass'. Normal stalks are not formed.	Resistant or moderately resistant varieties should be planted. Healthy seed (as mentioned under red rot) should be used. Setts may be treated with antibiotics like *Achromycin, Terramycin, Tylan, Erythromycin* @ 250 ppm.
5.	Mosaic	Sugarcane Mosaic Virus (SCMV)	Young leaves of the crown held against the light source display chlorotic and normal green area imparting mosaic pattern. The	Seed should be obtained from disease free plant crop. Secondary transmission of the disease by insect vectors can be controlled by application of Malathion

			chlorotic area may show reddening or necrosis. Leaf sheath may also display such sysmptoms.	(0.1%) or Dimecron(0.2%)

Annexure XXV

State wise number of sugar factories in operation in India

S. No.	State	2006-07	2007-08	2008-09	2009-10	2010-11	2011-12*
1.	Andhra Pradesh	38	38	35	35	37	37
2.	Assam	-	-	-	-	-	-
3.	Bihar	9	8	9	9	10	11
4.	Chhattisgarh	1	1	1	3	3	3
5.	Dadra Nagar Haveli	-	-	-	1	-	-
6.	Goa	1	1	1	1	1	1
7.	Gujrat	18	18	18	18	19	19
8.	Haryana	14	14	15	14	14	14
9.	Karnataka	47	51	50	54	59	58
10.	Kerala	-	-	-	-	-	-
11.	Madhya Pradesh	8	8	10	11	13	13
12.	Maharashtra	163	172	147	143	167	170
13.	Nagaland	-	-	-	-	-	-
14.	Orissa	5	6	5	4	5	5
15.	Puducherry	2	2	1	2	2	2
16.	Punjab	16	16	16	15	16	17
17.	Rajasthan	1	1	1	1	1	1
18.	Tamil Nadu	37	37	37	40	44	43
19.	Uttar Pradesh	133	132	132	128	125	124
20.	Uttarakhand	10	10	10	10	10	10
21.	West Bengal	1	1	1	1	1	1
	All India	504	516	489	490	527	529

2012. *Provisional*

Source: Cooperative sugar: Vol.44 (4), December 2012

State-wise sugar factories may be seen at Directorate website: http://dsd.dacnet.nic.in

Annexure XXVI

State wise duration of crushing season for sugarcane in India

(in days)

S. No.	State	2007-08	2008-09	2009-2010	2010-11	2011-12*
1	Andhra Pradesh	127	72	63	104	113
2	Bihar	93	59	66	87	97
3	Chhattisgarh	190	69	46	49	76
4	Gujrat	189	126	151	176	137
5	Haryana	150	63	59	109	146
6	Karnataka	177	93	127	166	149
7	Madhya Pradesh	116	47	55	80	73
8	Maharashtra	160	89	143	164	147
9	Orissa	75	46	43	71	90

10	Puducherry	179	113	75	141	206
11	Punjab	136	66	54	77	102
12	Rajasthan	103	51	60	60	39
13	Tamil Nadu	214	165	142	172	202
14	Uttar Pradesh	120	72	84	103	123
15	Uttarakhand	117	75	86	96	110
16	West Bengal	56	22	26	63	53
	All India	149	87	109	135	137

*Provisional, Source: Cooperative sugar: Vol.44 (4), December 2012

Annexure XXVII

STATE WISE AVERAGE SUGAR RECOVERY PERCENT CANE IN INDIA

S. No.	State	2006-07	2007-08	2008-09	2009-10	2010-11	2011-12*
1.	Andhra Pradesh	9.70	10.12	9.89	9.28	9.75	9.79
2.	Assam	-	-	-	-	-	-
3.	Bihar	8.67	9.24	9.04	9.49	9.30	9.45
4.	Chhattisgarh	7.62	9.00	8.43	7.82	8.72	8.21
5.	Dadra Nagar Haveli	-	-	-	2.44	-	-
6.	Goa	9.57	9.87	8.72	8.17	8.66	9.02
7.	Gujarat	10.64	10.67	10.72	10.53	9.99	10.61
8.	Haryana	9.75	9.87	9.05	9.37	9.01	9.10
9.	Karnataka	10.57	10.87	10.28	10.67	10.91	11.14
10.	Kerala	-	-	-	-	-	-
11.	Madhya Pradesh	9.29	9.54	9.62	9.40	9.66	9.69
12.	Maharashtra	11.39	11.92	11.44	11.51	11.29	11.65
13.	Nagaland	-	-	-	-	-	-
14.	Orissa	9.74	9.51	9.44	9.01	8.70	8.89
15.	Pondicherry	8.31	8.78	10.02	8.30	8.64	8.86
16.	Punjab	9.55	9.26	9.29	8.59	8.80	9.13
17.	Rajasthan**	9.04	7.05	8.82	7.80	7.73	7.66
18.	Tamil Nadu	9.25	9.32	9.62	8.94	9.09	9.35
19.	Uttar Pradesh	9.47	9.79	8.94	9.13	9.14	9.07
20.	Uttarakhand	9.54	9.74	9.20	9.19	9.35	9.10
21.	West Bengal	8.32	6.43	6.44	7.04	7.18	8.10
	All India	10.16	10.55	10.03	10.19	10.17	10.25

Provisional, ** includes recovery of sugar % from sugar beet

Source: Cooperative sugar: Vol.44 (4), December 2012

Annexure XXVIII

STATE WISE & SECTOR WISE INSTALLED ANNUAL SUGAR PRODUCTION CAPACITY AND UTILISATION OF CAPACITY DURING LAST FIVE YEARS

S. No.	State	2007-08		2008-09		2009-10		2010-11		2011-12*	
		Capacity (In lakh tones)	Utilisation of capacity (%)	Capacity (In lakh tones)	Utilisation of capacity (%)	Capacity (In lakh tones)	Utilisation of capacity (%)	Capacity (In lakh tones)	Utilisation of capacity (%)	Capacity (In lakh tones)	Utilisation of capacity (%)
1.	A.P.	9.3260	143.20	9.3260	63.60	9.3260	46.30	9.5710	105.10	1.05540	107.50
2.	Assam	0.1840	-	0.1840	-	0.1840	-	0.1840	-	-	-
3.	Bihar	4.8488	69.30	4.8488	44.20	4.8488	53.30	4.8488	79.40	4.8419	92.90
4.	Chhattisgarh	0.2230	171.10	0.2230	56.90	0.2230	38.70	0.6690	34.90	0.6900	51.80
5.	DN&H	-	-	-	-	-	-	0.1950	-	-	-
6.	Goa	0.0930	157.20	0.0930	101.10	0.0930	87.30	0.0930	135.70	0.1175	88.80
7.	Gujrat	10.7922	126.60	10.7922	93.80	10.7922	110.20	11.3034	109.30	12.3710	80.90
8.	Haryana	3.3572	111.80	5.7092	40.10	5.7092	43.50	5.7092	68.60	5.1480	95.90
9.	Karnataka	17.6570	164.30	19.4746	84.80	20.4844	124.90	21.1576	174.10	28.3540	136.60
10.	Kerala	0.1020	-	0.1020	-	0.1020	-	0.1020	-	-	-
11.	M.P.	1.6562	105.00	1.9907	30.20	2.5036	32.00	2.5036	65.60	2.7610	57.50
12.	Maharashtra	71.6109	126.70	72.6974	63.00	75.3032	93.80	78.3433	115.60	84.8280	105.80
13.	Nagaland	0.0640	-	0.0640	-	0.0640	-	0.0640	-	-	-
14.	Orissa	1.0179	62.30	1.0179	30.30	1.0179	22.20	1.0179	44.30	0.8268	78.60
15.	Puducherry	0.3830	132.00	0.3830	43.30	0.3830	48.60	0.3830	123.20	0.3930	162.30
16.	Punjab	6.8423	78.00	6.8423	35.40	6.8423	26.50	7.1913	42.00	4.7395	82.30
17.	Rajasthan	0.2320	25.50	0.2320	16.00	0.2320	16.30	0.2320	16.20	0.2335	9.40
18.	Tamil Nadu	16.9712	126.10	17.5214	91.20	19.0934	67.10	19.0934	96.70	23.1680	102.70
19.	Uttar Pradesh	73.2583	99.90	74.2543	54.70	74.7191	69.30	74.7191	78.80	66.0557	105.60
20.	Uttarakhand	4.1168	97.30	4.1168	54.10	4.1882	69.70	4.1882	72.20	3.2240	102.80
21.	West Bengal	0.0670	68.90	0.1414	13.40	0.1414	14.60	0.1414	35.30	0.1617	28.20
	All India	224.8028	117.20	230.0140	63.20	236.2507	80.00	241.7102	90.90	248.4676	106.00

**Provisional* Source: Cooperative sugar: Vol.44 (4), December 2012

Annexure XXIX

STATE WISE SUGAR PRODUCTION ('000 ton) DURING LAST 10 YEARS

Sl. No	State	2002-03	2003-04	2004-05	2005-06	2006-07	2007-08	2008-09*	2009-10	2010-11	2011-12*
1	**A.P.**	1210	886	982	1236	1680	1335	593	515	1006	1135
2	**Assam**	-	-	-	-	-	-	-	-	-	-
3	**Bihar**	408	274	253	422	451	336	214	258	385	450
4	**Chhattisgarh**	4	17	10	18	24	38	13	9	23	36
5	**Goa**	13	10	8	11	19	15	9	8	13	10
6	**Gujarat**	1252	1066	797	1168	1425	1366	1012	1189	1235	1000
7	**Haryana**	635	582	100	409	652	599	229	248	392	494
8	**Karnataka**	1868	1116	1040	1943	2662	2900	1654	2558	3683	3872
9	**Kerala**	2	-	-	-	-	-	-	-	-	-
10	**M.P.**	71	93	72	94	180	174	56	80	165	159
11	**Maharashtra**	6219	3175	2217	5197	9100	9075	4578	7067	9054	8977
12	**Nagaland**	-	-	-	-	-	-	-	-	-	-
13	**Orissa**	40	41	44	40	61	63	31	23	45	65
14	**Puducherry**	34	20	18	28	60	51	17	19	47	64
15	**Punjab**	586	390	315	338	486	534	242	181	302	390
16	**Rajasthan**	2	9	4	6	7	6	4	4	4	2
17	**Tamilnadu**	1644	921	1108	2142	2539	2141	1598	1280	1846	2379
18	**U.P.**	5651	4552	5037	5784	8475	7319	4064	5179	5887	6974
19	**Uttarakhand**	498	387	381	426	535	400	223	292	302	331
20	**W.B.**	8	7	5	5	8	5	2	2	5	5
	All India	**20145**	**13456**	**12691**	**19267**	**28364**	**26357**	**14539**	**18912**	**24394**	**26343**

*** Provisional**

Source: Cooperative sugar: Vol.44 (4), December 2012

Annexure-XXX

EXPORT- IMPORT OF SUGAR ON FINANCIAL YEAR BASIS FROM 2000-01 ONWARDS (AS PER DGCI & S, KOLKATA)

Financial year (April to March)	EXPORT		IMPORT	
	Quantity (Tonnes)	Value (Rs./ Crores)	Quantity (Tonnes)	Value (Rs./ Crores)
2000-01	338691	430.98	30404	31.11
2001-02	1456448	1728.29	26578	32.60
2002-03	1662370	1769.49	41430	32.83
2003-04	1200600	1216.59	74400	62.70
2004-05	108690	149.53	932740	976.18
2005-06	321204	569.11	558769	651.59
2006-07	1643403	3127.47	1052	3.49
2007-08	4684554	5412.16	496	2.24
2008-09	3331997	4448.74	386099	583.11
2009-10	44045	110.23	2424045	5961.24
2010-11	3249300	10352.27	1004100	2723.21
2011-12 (Up to September, 11)	2717300	8736.15	70454	209.64

Source: Cooperative Sugar, vol-44, No.4, Dec, 2012.

Annexure-XXXI

STATE WISE RATE OF CESS/ PURCHASE TAX ON SUGARCANE PAID BY SUGAR FACTORIES

S. No.	State	Cess/ Purchase tax in Rupees per tonne cane	
		2010-11	2011-12
1.	Andhra Pradesh	60.00	60.00
2.	Bihar	17.50	17.50
3.	Gujrat	2 % on cane price paid	2 % on cane price paid
4.	Haryana	15.00	15.00
5.	Karnataka	65.00 for factories with 10.50 % and above recovery + 10 road cess. 50.00 for factories with recovery below 10.50 % + 10 road cess.	65.00 for factories with 10.50 % and above recovery + 10 road cess. 50.00 for factories with recovery below 10.50 % + 10 road cess.
6.	Kerala	4.6 %	4.6 %
7.	Madhya Pradesh	4.5 % + 1 % Mandi tax	4.5 % + 1 % Mandi tax
8.	Maharashtra	3 % ad velorem	3 % ad velorem
9.	Punjab	5.5 % VAT + 5.00 cess	5.5 % VAT + 5.00 cess
10.	Rajasthan	2 % ad velorem	2 % ad velorem
11.	Tamil Nadu	60.00 + 5.00 cess	60.00 + 5.00 cess
12.	Uttar Pradesh	20.00	20.00
13.	Uttarakhand	20.00	20.00

Annexure-XXXII

Progress report of Sugarcane Development Scheme under Macro Management Mode on Agriculture (MMMA) for the year 2011-12

Rs. In Lakh

1	**Gujrat (MMMA)**	**March, 2012 (Rs. In Lakhs)**				
		Unit	**Target**		**Physical**	**Financial**
			Physical	**Financial**	**Achv.**	**Achv.**
1	Field demonstration	No	20	1.50	20	1.50
2	Seed Multiplication	Ha				
	a. Foundation nursery		100	4.00	100	4.00
	b. Primary nursery		200	4.00	196	3.92
3	Trainings	No				
	a. Farmers training		10	1.00	9	0.90
	b. State level training		2	0.40	0	0.00
4	Supply of drip irrigation infrastructure	Ha.	60	18.00	59.22	17.77
5	INM (supply of Organic manure)	Ha.	1600	64.00	1621	57.06
6	IPM (supply of Tricoderma culture)	Ha.	1600	14.40	1799	11.23
	Total			**107.30**		**96.38**
2	**Maharashtra**	**March, 2012 (Rs. In Lakhs)**				
A	**Approved Component**					
1	Field Demonstration	No	9500	712.50	10269	751.32
2	**Multiplication of Planting material (Seed Production Programme)**					
a	Foundation seed	ha	1800	72.00	1336	53.35
b	Certified seed		18000	360.00	16978	341.05
3	**Human Resource Develop.Prog.**					
a	Extension worker/officers training	No	38	7.60	38	7.77
b	Farmers visit	No	40	20.00	46	23.00
4	**Supply of Micronutrients**					
a	Distrubution of Micronutrients	Ha	27998	279.98	29612	287.45
b	Distrubution ofGypsum	Ha	25000	250.00	22003	218.79
c	Green mannuring /Decomposting Sugarcane Trash	Ha	9000	90.00	11437	94.31
5	**New initiatives**					
a	Farmer’s field School(FFS)	No	244	41.48	245	41.63
b	Tractor/ power drawn	No	1000	150.00	12980	121.75
	Total			**1983.56**		**1940.42**

3	**Odisha**	**March, 2012 (Rs. In Lakhs)**				
		Unit	**Target**		**Physical**	**Financial**
			Physical	**Financial**	**Achv.**	**Achv.**
1	Farmers Field School	No.	120	12.00	120	12.00
2	State level trainings	Ha.	1	0.20	1	0.20
3	Technology Demonstration	No.	1800	135.00	1800	135.00
4	Intercropping demonstration	No.	2064	10.32	2064	10.32
5	Seed Multiplication	Ac.	1000	8.00	1000	8.00
6	Distribution of seed cane	Ac.	3000	12.00	3000	12.00
	Total			**177.52**		**177.52**
4. Ulttrakhand			**March – 2012 (Rs in Lakhs)**			
1	Field Demonstration	Ha	260	2.25	15	2.25
2	Seed Multiplication	Ha	2220	0.69	17.25	0.69
3	Tractor Driven Implement	No	400	14.46	250	18.15
4	Bullock Driven Implement	No	1000	6.00	2035	5.92
5	State level Training	No	2	0.40	2	0.40
6	Farmer's level Training	No	12	1.20	12	1.20
	Total			**25.00**		**28.61**
Balance of last year Rs. 3.84 lakh (Rs. 25.00+3.84 = 28.84 Lakhs) is used in 2011-12						
5	**Chhattishgarh**	**March 2012 (Rs in Lakhs)**				
1	Demonstration of technology (0.5ha)	No.	180	13.50	164	11.67
2	Seed/ Planting material multiplication for foundation nursury	Ha.	25	1.00	4	0.16
3	Farmers training	No.	34	3.40	34	3.40
4	Visit of farmers to model farms, institute	No.	7	3.50	7	3.50
5	Contingencies & POL			1.60		1.50
	Total			**23.00**		**20.23**
6	**Karnataka**	**March 2012 (Rs in Lakhs)**				
1	Large scale demonstrations of technology	Ha	3000	225.00	6614	225.36
2	Demo on ratoon management	ha	2000	150.00	4859	156.30
3	Sugarcane demonstration using single eye bud.	Ha	200	15.00	33	2.48
4	Multiplication & distribution of Sugarcane sets.	Ha	500	10.00	387	7.80
	Total			**400.00**		**391.94**

7	MIZORUM		March 2012 (Rs in Lakhs)			
			Physical	Financial	Physical	Financial
		Unit	Target		Achv.	Achv.
1	Demonstration of technology	No.	60	4.5		4.5
2	Multiplication of planting material of HYV cane setts	No.	275	5.5		5.5
3	Trainings					
	a. Farmers training	No	26	2.6		2.6
	b. State level	No	2	0.4		0.4
4	Monitoring/ Inspection/ Visit preparation of report/ P.O.L. as contengency	No	4	2		2
	Total			**15**		**15**
8	**Bihar**	**March 2012 (Rs in Lakhs)**				
1	Farmers training	No.	600	60.00	551	54.85
2	Transfer of technology			40.50		39.50
	Total			**100.50**		**94.34**
9. Madhya Pradesh						
1.	Field Demonstration	No.	90	**3.75**	**28.00**	**1.99**
2	Seed Multiplication	Ha.	250	**5.00**	**47.00**	**0.86**
3	Drip Irrigation	Ha.	5	**1.50**	**1.00**	**0.30**
4	Micro nutrient subsidy	Ha.	1000	10	610.00	2.35
5	MHAT Plant	No.	2	6	2.00	6.00
6	Treatment of Planting material	Ha.	200	2	225.00	0.28
7	Farmers visit	No.	5	2.5	2.00	1.00
8	Tissue culture	No.	40000	0.5	0.00	0.00
9	Breeder seed	Ha.	2	0.5	1.00	0.25
10	Contingencies & video conference			8.25	0.00	2.32
	Total			**40.00**		**15.35**
	Grand Total			**2871.88**		**2779.79**

Based on Progress Report received from State department.

Annexure-XXXIII

STATEMENT SHOWING THE MINIMUM STATUTORY PRICE/ FAIR & REMUNERATIVE PRICE OF SUGARCANE FIXED BY THE GOVERNMENT

Year	Minimum Statutory price of Sugarcane (Rs. Per quintal)	Linked to basic sugar recovery % cane	Premium on every 0.1% increase in sugar recovery % cane (Rs. Per quintal)	Range of Minimum Sugarcane Price on the basis of Col. 1, 2 & 3 (Rs. Per quintal)
2000- 01	59.50	8.50	0.70	59.50 to 96.60
2001- 02	62.05	8.50	0.73	62.05 to 100.74
2002- 03	69.50	8.50	0.82	69.50 to 113.78
2003- 04	73.00	8.50	0.85	73.00 to 118.90
2004- 05	74.50	8.50	0.88	74.50 to 110.58
2005- 06	79.50	9.00	0.88	79.50 to 112.94
2006- 07	80.25	9.00	0.90	80.26 to 119.85
2007- 08	81.18	9.00	0.90	81.18 to 118.98
2008- 09	81.18	9.00	0.90	81.18 to 123.48
2009- 10	129.84 (FRP)	9.50	1.37	129.84 to 179.16
2010-11	139.12(FRP)	9.50	1.46	139.12 to 197.52
2011- 12	145.00 (FRP)	9.50	1.53	145.00 to 203.14
2012-13	170 (FRP)	9.50	1.79	NA

Source: Cooperative Sugar Vol. 44 (4) December 2012